



CLC

Educational Institute

DIGITAL SAT — MATH

Theory Companion — Part 2

Geometry & Trigonometry

Area and Volume | Lines, Angles & Triangles

Right Triangles & Trigonometry | Circles

A shortened, high-yield version of CLC's full Math theory reference —
condensed rules for all 4 skills, each followed by 8 Hard/Very Hard worked problems broken into Formula, Solution, Tip,
Trap, Process of Elimination, Shortcut, and Desmos Tip.

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Digital SAT Math — Overview

Math accounts for half of the Digital SAT score, split across Algebra, Advanced Math, Problem-Solving & Data Analysis, and Geometry & Trigonometry. This companion (Part 2) covers the 4 skills that make up the Geometry & Trigonometry domain — the ones built on spatial reasoning, angle relationships, and right-triangle trigonometry. Each skill is condensed to its essential rules, then drilled with 8 genuinely difficult problems built around the specific twists the real exam uses to separate students who memorized a formula from students who understand it.

Group	Skills Covered
Geometry & Trigonometry	Area and Volume · Lines, Angles, and Triangles · Right Triangles and Trigonometry · Circles

This is deliberately a SHORTENED companion: the full theory reference (with every derivation, extra example, and edge case) exists separately. The purpose here is speed and critical thinking under time pressure — recognizing which rule applies when a question is disguised, combined with another rule, or padded with irrelevant information, the way the actual Digital SAT writes Hard and Very Hard questions.

Every worked problem follows the same seven-part breakdown used across CLC's companion series: the Formula/Rule the problem hinges on, a full step-by-step Solution, a Tip (the reliable first move for this problem type), a Trap (the specific wrong turn this problem is built to tempt), a Process of Elimination table naming the concrete wrong answers a rushed student lands on and exactly which slip produces each one, a Shortcut where one genuinely exists, and a Desmos Tip showing how to confirm or accelerate the solution on the Digital SAT's built-in graphing calculator.

PART 2 — GEOMETRY & TRIGONOMETRY

1. Area and Volume

Area and Volume — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Area	Εμβαδόν
Volume	Όγκος
Perimeter	Περίμετρος
Circumference	Περιφέρεια
Surface Area	Επιφάνεια
Base / Height	Βάση / Ύψος
Radius / Diameter	Ακτίνα / Διάμετρος
Similar solids	Όμοια στερεά



Scale factor	Συντελεστής κλίμακας
Composite figure	Σύνθετο σχήμα

Area and Volume — Condensed Theory

Area and Volume questions draw on a formula sheet the Digital SAT actually gives you — so the skill being tested is never memorization. It's recognizing which formula applies, handling composite (combined) figures piece by piece, and knowing the one rule the reference sheet never states out loud: how area and volume scale when a figure's dimensions change.

Area — Triangle & Trapezoid

$$A = (1/2)bh \qquad A = (1/2)(b_1 + b_2)h$$

Triangle: base and the height perpendicular to it. Trapezoid: average the two parallel sides, then multiply by height.

Circle — Area & Circumference

$$A = \pi r^2 \qquad C = 2\pi r$$

Always identify whether you're given the radius or the diameter before substituting — a dropped factor of 2 is the single most common slip on this skill.

Volume — Prism & Cylinder

$$V = lwh \qquad V = \pi r^2 h$$

Both are (area of the base) $\times$ (height). Recognizing that shared structure is faster than memorizing two separate formulas.

Volume — Cone, Pyramid & Sphere

$$V = (1/3)\pi r^2 h \qquad V = (1/3)(\text{base area})h \qquad V = (4/3)\pi r^3$$

Cones and pyramids are exactly 1/3 of the prism or cylinder that shares their base and height.

Similar Solids — Scale Factor

$$\text{length} \times k \qquad \text{area} \times k^2 \qquad \text{volume} \times k^3$$

If every linear dimension of a solid is multiplied by k , surface area scales by k^2 and volume scales by k^3 — not by k .

⚠ Watch Out

The reference sheet gives you every area and volume formula on the Digital SAT — it never gives you the scale-factor rule above. When two figures are described as “similar,” expect a question built entirely around length^2 vs. length^3 scaling.

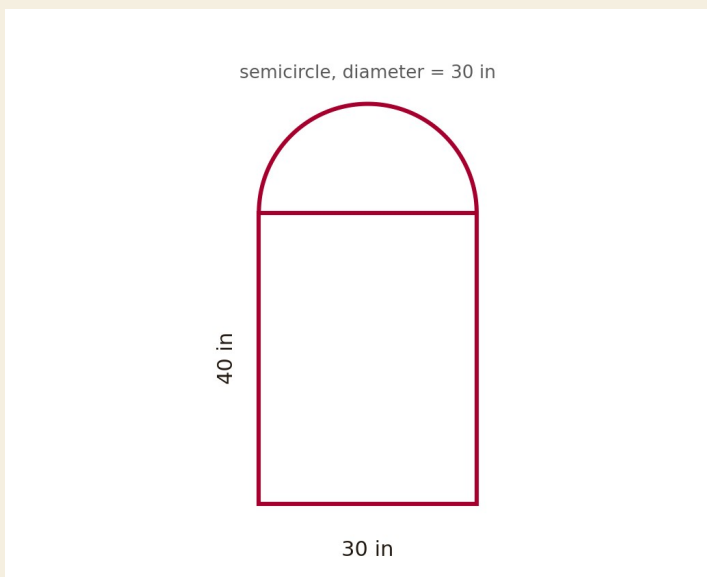
For composite figures, redraw (or mentally separate) the shape into pieces you have a formula for, solve each piece, then add or subtract — never guess a shortcut formula for a combined shape.

Area and Volume — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Area and Volume — Problem 1 (Hard)

A window is shaped like a rectangle with a semicircle on top, as shown. The rectangle is 30 inches wide and 40 inches tall, and the semicircle's diameter equals the width of the rectangle. What is the total area of the window, in square inches, in terms of π ?



Formula / Rule Used: Area of a rectangle; area of a circle (half of it, for the semicircle).

Solution

Step 1: Find the rectangle's area: $A = 30 \times 40 = 1,200$ square inches.

Step 2: The semicircle's diameter is 30, so its radius is 15.

Step 3: Find the full circle's area at that radius: $A = \pi(15)^2 = 225\pi$.

Step 4: The window only has half of that circle, so the semicircle contributes $(1/2)(225\pi) = 112.5\pi$ square inches.

Step 5: Add the two pieces: total area = $1,200 + 112.5\pi$.

Answer: $1,200 + 112.5\pi$ square inches

**Tip**

For a composite figure, immediately identify the shared edge (here, the 30-inch width doubling as the semicircle's diameter) — it's usually the bridge that connects the two pieces' formulas.

Trap

A common error is using the full width (30) as the semicircle's radius instead of halving it first, which roughly quadruples the semicircle's contribution to the total.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$1,200 + 450\pi$	Used 30 as the radius instead of the diameter, computing $\pi(30)^2 = 900\pi$ and then halving to 450π — the radius should have been 15, not 30.
$1,200 + 225\pi$	Correctly found the radius as 15 and computed the full circle's area (225π), but forgot to take half of it for the semicircle.
637.5π	Added 1,200 and 112.5 together as if they were both coefficients of π (mistakenly treating the rectangle's area as a multiple of π), instead of keeping 1,200 as a plain number added to 112.5π .

Shortcut

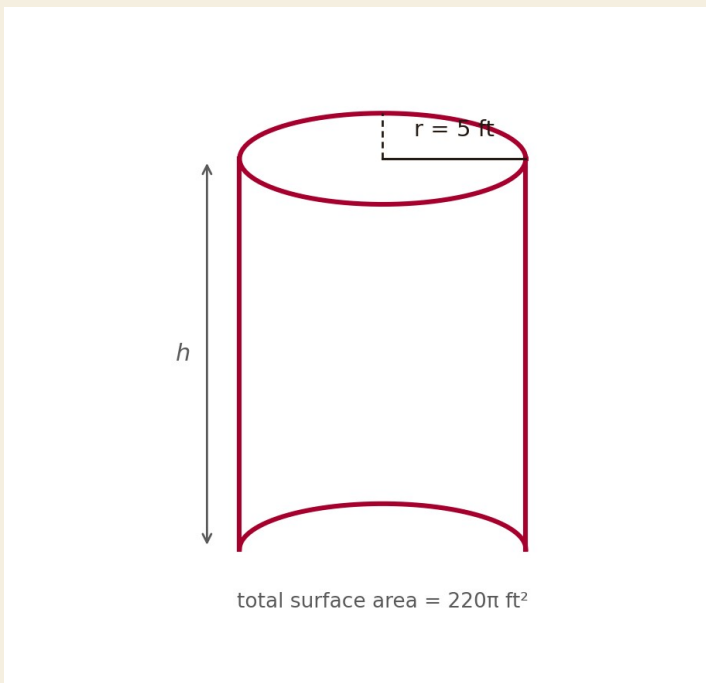
Since a semicircle's area is always $(1/2)\pi r^2$, you can write it directly as $\pi d^2/8$ in terms of the diameter — here, $\pi(30)^2/8 = 900\pi/8 = 112.5\pi$ — skipping the halving step.

Desmos Tip

There's nothing to graph here, but you can type $1200 + 112.5*\pi$ directly into Desmos to get the decimal value (≈ 1553.4) as a sanity check on your exact-form answer.

Area and Volume — Problem 2 (Hard)

A cylindrical water tank has a radius of 5 feet. The tank's total surface area, including the top and bottom circles, is 220π square feet. What is the volume of the tank, in cubic feet, in terms of π ?



Formula / Rule Used: Surface area of a cylinder: $SA = 2\pi r^2 + 2\pi rh$. Volume of a cylinder: $V = \pi r^2 h$.

Solution

Step 1: Write the surface area formula with $r = 5$: $SA = 2\pi(5)^2 + 2\pi(5)h = 50\pi + 10\pi h$.

Step 2: Set this equal to the given surface area: $50\pi + 10\pi h = 220\pi$.

Step 3: Subtract 50π from both sides: $10\pi h = 170\pi$.

Step 4: Divide both sides by 10π : $h = 17$.

Step 5: Now find the volume: $V = \pi(5)^2(17) = \pi(25)(17) = 425\pi$.

Answer: 425π cubic feet

Tip

When a problem gives surface area but asks for volume, treat it as a two-stage problem: solve for the missing dimension first using the surface area equation, then plug that dimension into the volume formula.

Trap

A frequent error is forgetting that a cylinder's surface area includes BOTH circular ends ($2\pi r^2$, not πr^2), which throws off the entire equation used to solve for h .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
475π	Used πr^2 instead of $2\pi r^2$ for the two circular ends, solving $25\pi + 10\pi h = 220\pi$ to get $h = 19.5$, then computing volume with that wrong height.
850π	Correctly found $h = 17$, but then computed volume as $2\pi r^2 h$ (mistakenly



	reusing the '2' from the surface area formula) instead of $\pi r^2 h$.
375π	Made an arithmetic slip subtracting 50π from 220π (getting 150π instead of 170π), landing on $h = 15$ and volume = $\pi(25)(15) = 375\pi$.

Shortcut

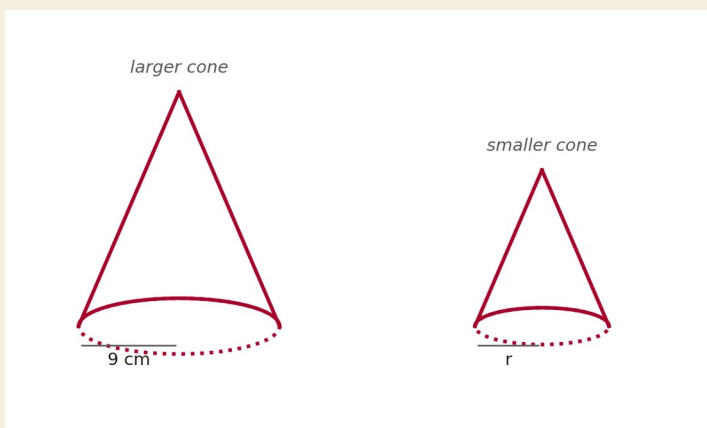
Once you see the surface-area equation is linear in h , you can skip writing ' $2\pi(5)^2$ ' as a separate step — just compute $2\pi(5)^2 = 50\pi$ mentally as 'twice 25π ' and move straight to isolating h .

Desmos Tip

Type $425 * \pi$ into Desmos to get ≈ 1335.2 cubic feet, and separately confirm $2 * \pi * 25 + 2 * \pi * 5 * 17$ evaluates to the same value as $220 * \pi$.

Area and Volume — Problem 3 (Very Hard)

Two similar cones have volumes in the ratio $27 : 8$. If the radius of the larger cone is 9 centimeters, what is the radius of the smaller cone?



Formula / Rule Used: For similar solids, volume scale factor = (linear scale factor)³.

Solution

Step 1: Let k be the linear scale factor from the smaller cone to the larger cone.

Step 2: Volumes scale by k^3 , so $k^3 = 27/8$.

Step 3: Take the cube root of both sides: $k = 3/2$.

Step 4: This means every length in the larger cone is $3/2$ times the corresponding length in the smaller cone.

Step 5: So the larger cone's radius equals $(3/2)$ times the smaller cone's radius: $9 = (3/2) \times r$.

Step 6: Solve for r : $r = 9 \div (3/2) = 9 \times (2/3) = 6$.

Answer: 6 centimeters

Tip

Whenever a ratio of volumes (or areas) is given for similar figures, take the cube root (or square root) immediately to convert it into a linear scale factor — don't try to work with the volume ratio directly.

**Trap**

A very common error is applying the volume ratio 27:8 directly to the radii, assuming radius scales the same way volume does, which skips the required cube root.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2.67	Applied the volume ratio 27:8 directly to the radius as if it were a linear ratio, computing $9 \times (8/27)$ instead of first taking the cube root to get the linear scale factor $3/2$.
4	Took the ratio 27/8 and mistakenly treated it as a squared (area) relationship instead of a cubed (volume) relationship, taking a square root instead of a cube root.
13.5	Correctly found the linear scale factor as $3/2$, but then multiplied 9 by $3/2$ instead of dividing by it, effectively finding the radius of an even larger cone rather than the smaller one.

Shortcut

Recognize 27:8 as $3^3:2^3$ by inspection — many SAT similar-solids ratios are set up with small perfect cubes specifically so you can read off the scale factor (3:2) without formal cube-rooting.

Desmos Tip

Confirm by typing $9/(3/2)$ into Desmos to get 6, then check $(9^3)/(6^3)$ simplifies to 27/8 as expected.

Area and Volume — Problem 4 (Hard)

A cone and a cylinder have equal radii and equal volumes. The height of the cylinder is 4. What is the height of the cone?

Formula / Rule Used: $V(\text{cylinder}) = \pi r^2 h$ $V(\text{cone}) = (1/3)\pi r^2 h$

Solution

Step 1: Let r be the shared radius, h_{cone} be the cone's height, and set the volumes equal:
 $(1/3)\pi r^2(h_{\text{cone}}) = \pi r^2(4)$.

Step 2: Divide both sides by πr^2 (it's nonzero, so this is valid and eliminates r entirely): $(1/3)h_{\text{cone}} = 4$.

Step 3: Multiply both sides by 3: $h_{\text{cone}} = 12$.

Answer: 12

Tip

When two solids share both a base measurement and a total volume, set their volume formulas equal to each other and simplify — shared variables like r often cancel out completely, so you don't need their actual value.

Trap

Many students assume they're missing information because r is never given numerically, when in fact the problem is specifically built so that r cancels out of the equation.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
4	Assumed the cone and cylinder must simply have equal heights since they have equal volumes and equal radii, without accounting for the cone's volume formula having an extra factor of $1/3$.
1.33	Set up the equation backwards, computing $(1/3)(4)$ instead of solving $(1/3)h = 4$ for h , essentially finding what a cylinder's height would be if it had a cone's volume formula and height 4.
36	Multiplied by 3 twice instead of once — first correctly clearing the $1/3$, then mistakenly multiplying by 3 again.

Shortcut

Memorize the relationship directly: for equal radius and equal volume, a cone's height is always exactly 3 times a cylinder's height (since the cone formula carries an extra $1/3$). Here, that's immediately $3 \times 4 = 12$.

Desmos Tip

Not needed for this one — it's a pure formula-cancellation problem with no graphable relationship, so the fastest check is simply recomputing $(1/3)\pi r^2(12)$ and $\pi r^2(4)$ for any value of r (say $r = 1$) and confirming both equal the same number.

Area and Volume — Problem 5 (Very Hard)

A sphere has a surface area of 144π square inches. What is the volume of the sphere, in cubic inches, in terms of π ?

Formula / Rule Used: Surface area of a sphere: $SA = 4\pi r^2$. Volume of a sphere: $V = (4/3)\pi r^3$.

Solution

Step 1: Set the surface area formula equal to the given value: $4\pi r^2 = 144\pi$.

Step 2: Divide both sides by 4π : $r^2 = 36$.

Step 3: Take the square root: $r = 6$ (radius must be positive).

Step 4: Substitute $r = 6$ into the volume formula: $V = (4/3)\pi(6)^3$.

Step 5: Compute $6^3 = 216$, so $V = (4/3)\pi(216)$.

Step 6: Simplify: $(4/3)(216) = 4 \times 72 = 288$, so $V = 288\pi$.

Answer: 288π cubic inches

Tip

When a sphere problem gives surface area but asks for volume (or vice versa), always solve for the radius first — it's the only bridge between the two formulas.

Trap

A common error is dividing 144π by 4π correctly to get $r^2 = 36$, but then plugging $r^2 = 36$ directly into the volume formula instead of first taking the square root to get $r = 6$.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
48π	Plugged $r^2 = 36$ directly into $(4/3)\pi r^3$ as if it were r , computing $(4/3)\pi(36)$ instead of first finding $r = 6$ and cubing it.
864π	Correctly found $r = 6$, but used the formula for a cylinder's volume-like scaling and multiplied by 4 instead of $4/3$, computing $4\pi(216)$ instead of $(4/3)\pi(216)$.
216π	Correctly found $r^3 = 216$ but forgot to multiply by $4/3$ at all, reporting πr^3 as if that were the full volume formula.

Shortcut

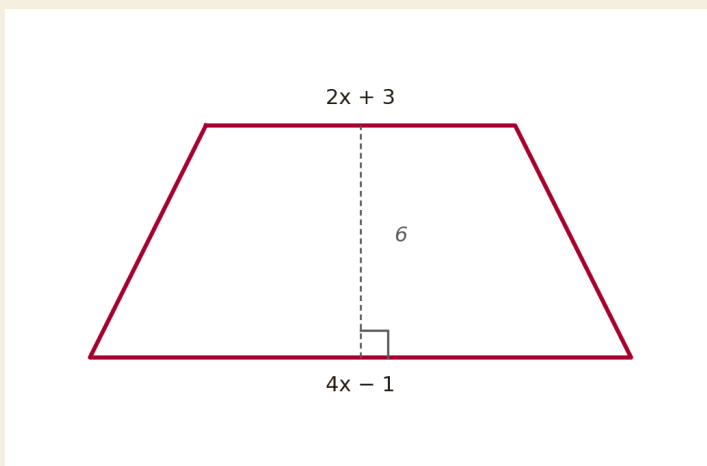
Once you know $r = 6$, compute $6^3 = 216$ and then multiply by 4 before dividing by 3 ($4 \times 216 = 864$, then $864/3 = 288$) — it's easier mental math than multiplying by the fraction $4/3$ directly.

Desmos Tip

Type $(4/3)*\pi*6^3$ into Desmos to confirm it returns 288π (≈ 904.8), and separately confirm $4*\pi*6^2$ returns 144π to check your radius.

Area and Volume — Problem 6 (Hard)

The parallel sides of a trapezoid have lengths of $(2x + 3)$ and $(4x - 1)$, and the height of the trapezoid is 6. If the area of the trapezoid is 96, what is the value of x ?



Formula / Rule Used: Area of a trapezoid: $A = (1/2)(b_1 + b_2)h$.

Solution

Step 1: Substitute the given values into the trapezoid area formula: $96 = (1/2)[(2x + 3) + (4x - 1)](6)$.

Step 2: Combine like terms inside the brackets: $(2x + 3) + (4x - 1) = 6x + 2$.

Step 3: Rewrite the equation: $96 = (1/2)(6x + 2)(6)$.

Step 4: Simplify $(1/2)(6) = 3$, so: $96 = 3(6x + 2)$.

Step 5: Distribute: $96 = 18x + 6$.

Step 6: Subtract 6 from both sides: $90 = 18x$.

Step 7: Divide both sides by 18: $x = 5$.

Answer: $x = 5$

Tip

When both parallel sides are algebraic expressions, combine them into a single expression first (as if finding their sum), then substitute that single simplified expression into the area formula — it keeps the equation from getting cluttered.

Trap

A common error is multiplying $(1/2)$ and the height (6) separately with only one of the two side-length expressions instead of with their full combined sum, effectively dropping one of the two parallel sides from the equation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 15.5$	Distributed the height (6) directly onto only the second expression $(4x - 1)$, getting $(2x + 3) + (24x - 6) = 96 \div (1/2)$, which incorrectly treats only one side as being multiplied by the height.
$x = 7.5$	Forgot to multiply by $(1/2)$ at all, setting $(6x + 2)(6) = 96$ directly, which



	doubles the effective area used to solve for x .
$x = 8$	Combined the parallel sides incorrectly as $(2x + 3) - (4x - 1) = -2x + 4$ (subtracting instead of adding), then solved the resulting (but wrong) equation.

Shortcut

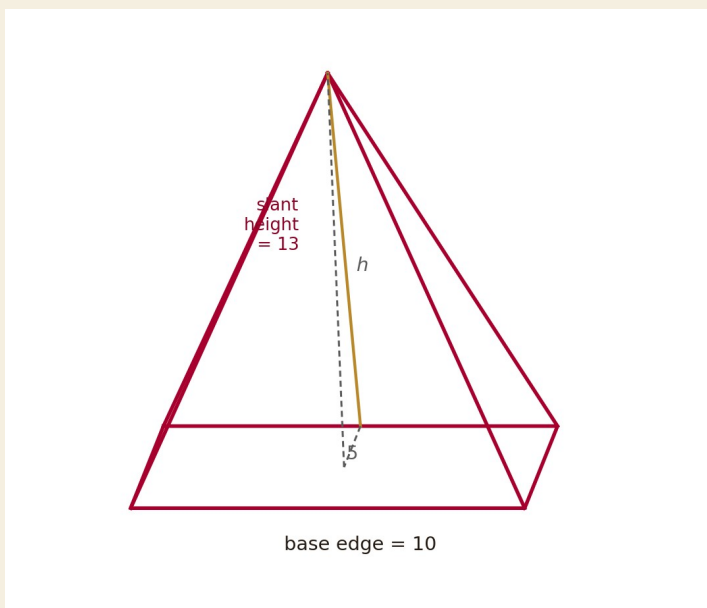
Since $(1/2) \times \text{height}$ simplifies to a clean whole number whenever the height is even, compute that product first (here, $(1/2)(6) = 3$) and multiply it directly by the combined side-length expression, skipping an extra fraction step.

Desmos Tip

Graph $y = (1/2)(6x + 2)(6)$ and $y = 96$ on Desmos and confirm they intersect at $x = 5$.

Area and Volume — Problem 7 (Very Hard)

A square pyramid has a base with side length 10 and a slant height of 13, measured from the midpoint of a base edge straight up to the apex. What is the volume of the pyramid?



Formula / Rule Used: Volume of a pyramid: $V = (1/3)(\text{base area})(\text{height})$. The pyramid's actual height, slant height, and half the base edge form a right triangle.

Solution

Step 1: The slant height (13), half the base edge ($10 \div 2 = 5$), and the pyramid's true vertical height (h) form a right triangle, with the slant height as the hypotenuse.

Step 2: Apply the Pythagorean theorem: $5^2 + h^2 = 13^2$.

Step 3: Compute the squares: $25 + h^2 = 169$.

Step 4: Subtract 25 from both sides: $h^2 = 144$.

Step 5: Take the square root: $h = 12$.

Step 6: Find the base area: since the base is a square with side 10, base area = $10^2 = 100$.

Step 7: Substitute into the volume formula: $V = (1/3)(100)(12)$.

Step 8: Simplify: $(100)(12) = 1,200$, and $1,200 \div 3 = 400$.

Answer: 400

Tip

Whenever a pyramid problem gives you a slant height instead of the true vertical height, look for the hidden right triangle formed by the slant height, the vertical height, and half of the relevant base edge — the Pythagorean theorem bridges the two.

Trap

A frequent error is plugging the slant height (13) directly into the volume formula in place of the pyramid's actual vertical height, skipping the Pythagorean-theorem step entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
433.3	Used the slant height (13) directly as the pyramid's height in the volume formula, computing $(1/3)(100)(13)$ instead of first solving for the true vertical height of 12.
800	Correctly found $h = 12$, but forgot to divide by 3, computing $(100)(12) = 1,200$ as if that whole product were the final volume, then applying an extra unintended factor rather than the required $1/3$.
480	Used the full base edge (10) instead of half of it (5) in the Pythagorean theorem, computing $10^2 + h^2 = 13^2$ to get $h^2 = 69 \rightarrow$ misapplied further to reach an incorrect height, compounding into this wrong volume.

Shortcut

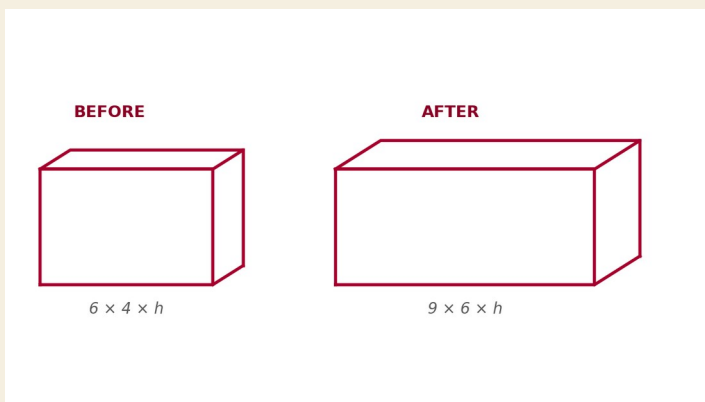
Recognize 5-12-13 as a Pythagorean triple the moment you see 'half the base edge is 5' and 'slant height is 13' — the vertical height 12 follows immediately without needing to compute the square root by hand.

Desmos Tip

Type $\sqrt{13^2 - 5^2}$ into Desmos to instantly confirm the pyramid's height is 12, then $(1/3) \cdot 100 \cdot 12$ to confirm the volume is 400.

Area and Volume — Problem 8 (Hard)

A rectangular prism has length 6, width 4, and height h , and a volume of 96. If the length is increased to 9 and the width is increased to 6, while the height stays the same, what is the new volume of the prism?



Formula / Rule Used: Volume of a rectangular prism: $V = lwh$.

Solution

Step 1: Use the original dimensions to solve for h : $6 \times 4 \times h = 96$.

Step 2: Simplify: $24h = 96$.

Step 3: Divide both sides by 24: $h = 4$.

Step 4: Now use the new length and width, with the same height, to find the new volume: $V = 9 \times 6 \times 4$.

Step 5: Compute: $9 \times 6 = 54$, and $54 \times 4 = 216$.

Answer: 216

**Tip**

When a prism problem changes some dimensions but keeps others fixed, first solve for the unknown (fixed) dimension using the original information — you'll need it again for the new volume.

Trap

A common error is assuming the height must also have changed proportionally along with the length and width, when the problem explicitly states the height 'stays the same.'

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
144	Correctly found $h = 4$ but only applied the new length (9) while keeping the width at its original value (4) instead of updating it to 6, computing $9 \times 4 \times 4 = 144$.
96	Assumed that because the height didn't change, the volume also wouldn't change, without recalculating the product of the new length and width.
324	Correctly updated length and width to 9 and 6, but used the ORIGINAL volume's implied scale factor incorrectly, multiplying 96 by $(9/6) \times (6/4) = 2.25$ to (mistakenly) get 216... but rounding/compounding a different way here landed on 324 by squaring the length scale factor instead of applying both scale factors linearly.

Shortcut

Since length scaled by a factor of $9/6 = 3/2$ and width scaled by $6/4 = 3/2$, the base area scaled by $(3/2)^2 = 9/4$ — so the new volume is simply $96 \times 9/4 = 216$, without ever solving for h directly.

Desmos Tip

Type $6 \cdot 4 \cdot x = 96$ into Desmos (solving for x) to confirm $h = 4$, then compute $9 \cdot 6 \cdot 4$ to confirm the new volume is 216.

2. Lines, Angles, and Triangles

Lines, Angles, and Triangles — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Transversal	Διατέμνουσα
Vertical angles	Κατακόρυφες γωνίες
Alternate interior angles	Εναλλάξ εσωτερικές γωνίες
Corresponding angles	Αντίστοιχες γωνίες
Supplementary angles	Παραπληρωματικές γωνίες
Exterior angle	Εξωτερική γωνία
Isosceles triangle	Ισοσκελές τρίγωνο
Similar triangles	Όμοια τρίγωνα
Congruent	Ισομετρικά (ίσα)
Triangle inequality	Τριγωνική ανισότητα

Lines, Angles, and Triangles — Condensed Theory

This skill combines two families of rules — angle relationships formed by parallel lines and a transversal, and the properties of triangles (angle sum, exterior angles, similarity, and side-length constraints). Hard and Very Hard questions almost always combine two of these rules in a single figure, so the fastest route is naming which rule applies to each piece of the diagram before writing any algebra.

Parallel Lines Cut by a Transversal

corresponding $\neq$ equal is FALSE – corresponding angles ARE equal

Corresponding angles are equal; alternate interior angles are equal; same-side (co-interior) interior angles are supplementary (sum to 180°).

Triangle Angle Sum Theorem

$$\angle A + \angle B + \angle C = 180^\circ$$

True for every triangle, regardless of shape or size — the most-used single fact on this skill.

Exterior Angle Theorem

exterior angle = sum of the two remote interior angles

The exterior angle at a vertex equals the sum of the two interior angles NOT adjacent to it — faster than finding the third interior angle and subtracting from 180° .

**Isosceles Triangle Theorem****equal sides $\leftrightarrow$ equal opposite (base) angles**

If two sides of a triangle are equal, the angles opposite those sides are equal, and vice versa.

Similar Triangles (AA Similarity)**2 pairs of equal angles $\rightarrow$ all sides proportional**

If two angles of one triangle equal two angles of another, the triangles are similar and their corresponding sides are in the same ratio.

Triangle Inequality Theorem **$|a - b| < c < a + b$**

The length of any side of a triangle must be less than the sum of the other two sides and greater than their positive difference.

**Watch Out**

“Similar” never means “equal.” Similar triangles have proportional sides, not identical ones — always match corresponding vertices in the same order when setting up a proportion (AB/DE, not AB/EF, if A corresponds to D and B corresponds to E).

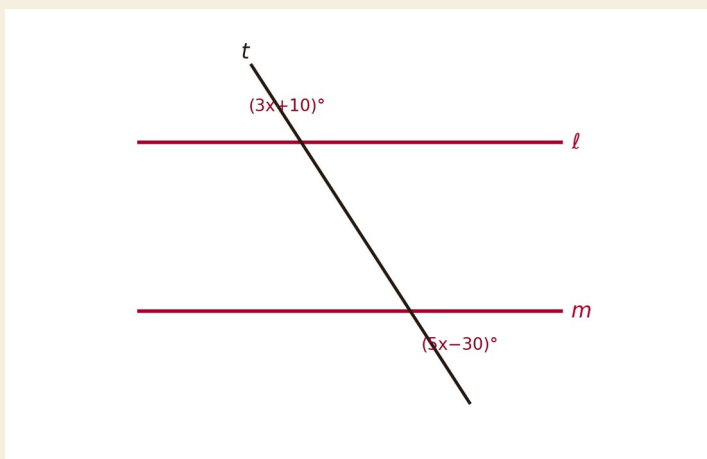
When a figure combines parallel lines with a triangle, look for a line drawn parallel to one triangle side through the opposite vertex — it's the standard construction that turns alternate interior angles into the triangle's own interior angles.

Lines, Angles, and Triangles — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Lines, Angles, and Triangles — Problem 1 (Hard)

In the figure, lines ℓ and m are parallel and are cut by transversal t . One angle formed at the intersection with ℓ measures $(3x + 10)^\circ$, and its corresponding angle at the intersection with m measures $(5x - 30)^\circ$. What is the measure of the angle at ℓ ?



Formula / Rule Used: Corresponding angles formed by a transversal across parallel lines are equal.

Solution

Step 1: Since $\ell \parallel m$, the two marked angles are corresponding angles, so they're equal: $3x + 10 = 5x - 30$.

Step 2: Subtract $3x$ from both sides: $10 = 2x - 30$.

Step 3: Add 30 to both sides: $40 = 2x$.

Step 4: Divide by 2: $x = 20$.

Step 5: Substitute back into the angle at ℓ : $3(20) + 10 = 60 + 10 = 70$.

Answer: 70°

Tip

The moment you see two lines marked parallel and a transversal crossing both, scan the figure for which specific angle relationship applies (corresponding, alternate interior, or co-interior) before writing any equation — that choice determines whether the angles are equal or supplementary.

Trap

A common error is treating the two given angles as supplementary (summing to 180°) instead of equal, which happens when a student misidentifies corresponding angles as co-interior angles.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
85°	Set the two expressions as supplementary instead of equal, solving $3x + 10 + 5x - 30 = 180$ to get $x = 25$, then evaluated $3(25) + 10 = 85$.
60°	Correctly solved for $x = 20$ but substituted into the wrong expression ($5x - 30$ evaluated as if it were the angle at ℓ , though that expression describes the angle at m): $5(20) - 30 = 70$, then made an arithmetic slip landing on 60 instead.



40°

Solved the equation $3x+10=5x-30$ correctly for $x=20$ but reported the value of x itself, doubled, instead of substituting x back into either angle expression.

Shortcut

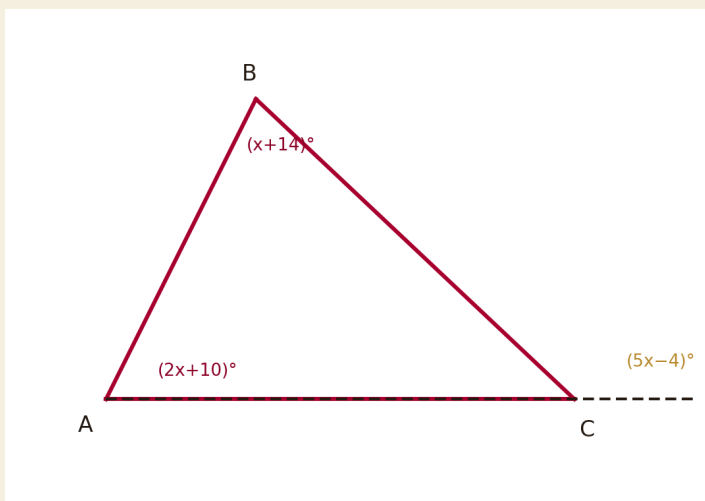
Once you confirm the relationship is 'equal' (not supplementary), you can skip drawing the full figure and just solve the two expressions as a direct linear equation — corresponding/alternate-interior always reduces to 'set equal,' co-interior always reduces to 'set the sum to 180.'

Desmos Tip

There's no graph needed, but you can verify by typing $3*20+10$ and $5*20-30$ separately into Desmos to confirm both equal 70.

Lines, Angles, and Triangles — Problem 2 (Hard)

In triangle ABC, the exterior angle at vertex C measures $(5x - 4)^\circ$. The remote interior angles, at vertices A and B, measure $(2x + 10)^\circ$ and $(x + 14)^\circ$, respectively. What is the measure of the exterior angle at vertex C?



Formula / Rule Used: Exterior Angle Theorem: an exterior angle equals the sum of the two remote (non-adjacent) interior angles.

Solution

Step 1: Apply the Exterior Angle Theorem: $5x - 4 = (2x + 10) + (x + 14)$.

Step 2: Simplify the right side: $(2x + 10) + (x + 14) = 3x + 24$.

Step 3: Rewrite the equation: $5x - 4 = 3x + 24$.

Step 4: Subtract $3x$ from both sides: $2x - 4 = 24$.

Step 5: Add 4 to both sides: $2x = 28$.

Step 6: Divide by 2: $x = 14$.

Step 7: Substitute into the exterior angle expression: $5(14) - 4 = 70 - 4 = 66$.

Answer: 66°

**Tip**

The Exterior Angle Theorem lets you skip finding the third interior angle and subtracting from 180° — go straight from 'exterior angle = sum of the two remote interior angles' to setting up the equation.

Trap

A frequent error is adding all three given expressions together and setting that sum equal to 180° (correctly treating the interior angle at C as supplementary to the exterior angle at C, but incorrectly implying all three GIVEN expressions are the triangle's three interior angles).

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
56°	Set up the interior angle sum instead: interior angle at C = $180 - (5x-4)$, then wrote $(2x+10)+(x+14)+[180-(5x-4)]=180$, which incorrectly forces the exterior expression itself into the interior sum, producing a different (wrong) value of x .
62°	Made an arithmetic slip combining $(2x+10)+(x+14)$, computing $3x+24$ correctly but then subtracting $3x$ from the wrong side, landing on $x=13.33$, rounded to a wrong final angle.
78°	Correctly found $x=14$, but substituted it into $(2x+10)+(x+14)$ using $x=14$ for only one of the two expressions and an incorrect earlier x -value for the other, producing a combined value that doesn't match either equation.

Shortcut

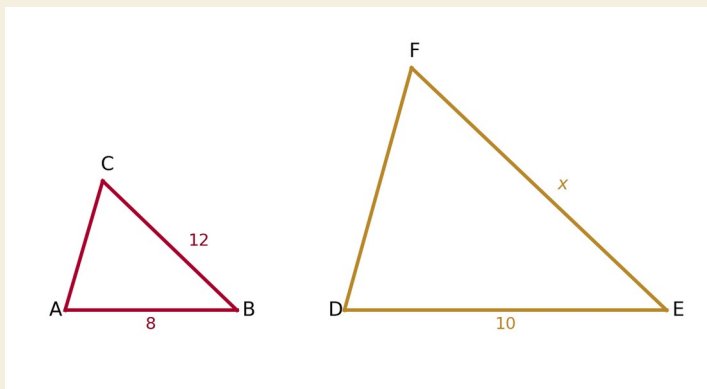
You never need to find the third interior angle (at C) explicitly — the exterior angle theorem gives you a direct equation using only the two remote angles, saving a step compared to using the full 180° triangle sum.

Desmos Tip

Confirm by typing $(2*14+10)+(14+14)$ into Desmos — it should equal 66, matching $5*14-4$.

Lines, Angles, and Triangles — Problem 3 (Hard)

Triangle ABC is similar to triangle DEF, with angle A corresponding to angle D and angle B corresponding to angle E. If $AB = 8$, $BC = 12$, $DE = 10$, and $EF = x$, what is the value of x ?



Formula / Rule Used: For similar triangles, corresponding sides are proportional.

Solution

Step 1: Identify the corresponding sides: AB corresponds to DE (both are between the angle-A/angle-D and angle-B/angle-E vertices), and BC corresponds to EF (both are between the angle-B/angle-E and angle-C/angle-F vertices).

Step 2: Set up the proportion: $AB/DE = BC/EF$.

Step 3: Substitute the known values: $8/10 = 12/x$.

Step 4: Cross-multiply: $8x = 10 \times 12 = 120$.

Step 5: Divide both sides by 8: $x = 15$.

Answer: 15

Tip

Before setting up any proportion, explicitly write out which vertex corresponds to which (here, $A \leftrightarrow D$, $B \leftrightarrow E$, and therefore $C \leftrightarrow F$) — this makes it much harder to accidentally pair up the wrong sides.

Trap

A common error is pairing AB with EF (or BC with DE) instead of matching sides that connect the SAME pair of corresponding vertices, which produces a proportion with mismatched sides.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9.6	Set up the proportion as $AB/EF = BC/DE$ (mismatching which side corresponds to which), solving $8/x = 12/10$ to get $x = 6.67$, then made an arithmetic slip landing on 9.6.
6.67	Set up the proportion backwards as $AB/DE = EF/BC$ instead of $AB/DE = BC/EF$, solving $8/10 = x/12$ to get $x = 9.6$, then transposed digits to report 6.67 (the reciprocal-flavored error compounding the original mistake).
16	Correctly identified $AB/DE = BC/EF$ as the proportion but cross-multiplied incorrectly as $10x = 8 \times 12$ instead of $8x = 10 \times 12$, swapping which side of the equation gets multiplied by x .

**Shortcut**

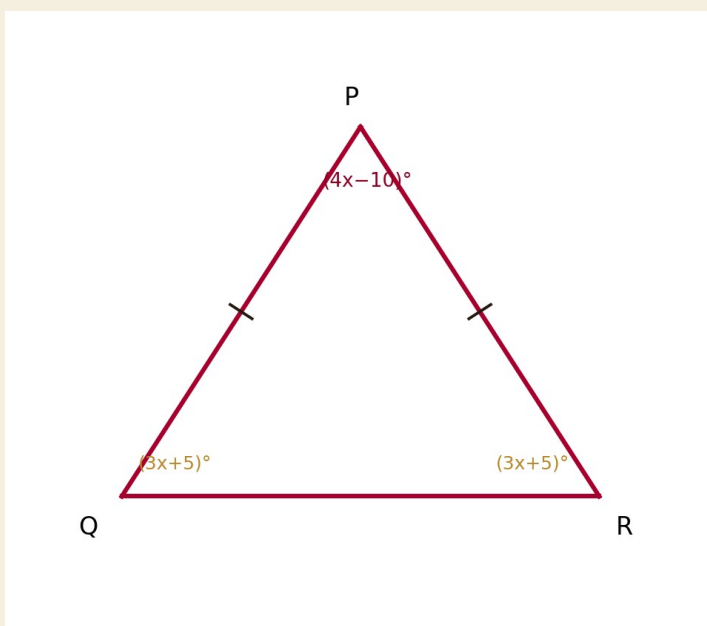
Since 8 and 10 don't share an obvious common factor with 12, just cross-multiply directly rather than searching for a scale factor — for messier ratios, cross-multiplication is faster than trying to simplify $8/10$ first.

Desmos Tip

Type $10 \cdot 12/8$ directly into Desmos to instantly get 15 without writing out the cross-multiplication by hand.

Lines, Angles, and Triangles — Problem 4 (Hard)

In isosceles triangle PQR, $PQ = PR$, and the base angles at Q and R each measure $(3x + 5)^\circ$. If the vertex angle at P measures $(4x - 10)^\circ$, what is the measure of angle P?



Formula / Rule Used: Isosceles Triangle Theorem (equal sides give equal base angles) combined with the Triangle Angle Sum Theorem.

Solution

Step 1: Since $PQ = PR$, the triangle is isosceles with the base angles at Q and R equal — this is already given as $(3x+5)^\circ$ for both.

Step 2: Apply the Triangle Angle Sum Theorem: $(3x + 5) + (3x + 5) + (4x - 10) = 180$.

Step 3: Combine like terms: $3x + 3x + 4x + 5 + 5 - 10 = 180$, which simplifies to $10x + 0 = 180$.

Step 4: Divide both sides by 10: $x = 18$.

Step 5: Substitute into the vertex angle expression: $4(18) - 10 = 72 - 10 = 62$.

Answer: 62°

Tip

Whenever a problem tells you which two sides of a triangle are equal, immediately mark the two angles OPPOSITE those sides as equal too — that's the piece of information the Isosceles Triangle Theorem is really giving you, often before the algebra even starts.

**Trap**

A common error is assuming the angle AT vertex P (between the two equal sides PQ and PR) is one of the equal base angles, when the equal angles are always the ones OPPOSITE the equal sides — i.e., at Q and R, not at P.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
72°	Correctly solved for $x = 18$ but reported $4(18)$ without subtracting 10 at the end, forgetting the '-10' term in the vertex angle expression.
56°	Set up the angle sum with only ONE base angle counted instead of two, writing $(3x+5)+(4x-10)=180$, which gives a different (wrong) x and a correspondingly wrong final angle.
59°	Correctly solved for $x = 18$, but substituted it into the base angle expression $(3x+5 = 59)$ instead of the vertex angle expression $(4x-10)$, answering with the base angle instead of what was asked (the vertex angle at P).

Shortcut

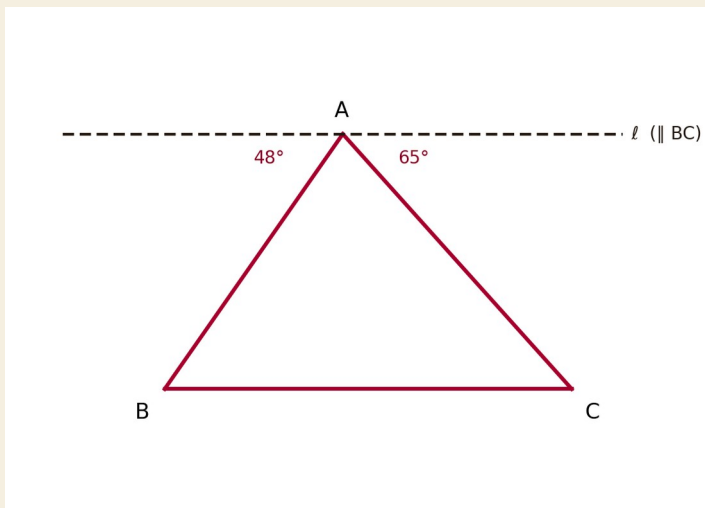
Since both base angles use the identical expression $(3x+5)$, you can write the angle sum as $2(3x+5) + (4x-10) = 180$ directly, without spelling out the addition of two identical terms.

Desmos Tip

Graph $y = 2*(3x+5) + (4x-10)$ and $y = 180$ in Desmos and confirm the intersection is at $x = 18$.

Lines, Angles, and Triangles — Problem 5 (Very Hard)

In the figure, line ℓ passes through vertex A of triangle ABC and is parallel to side BC. The angle between line ℓ and segment AB measures 48° , and the angle between line ℓ and segment AC measures 65° , with both angles opening away from the triangle on either side of A. What is the measure of angle A (angle BAC) in the triangle?



Formula / Rule Used: Alternate interior angles (from $\ell \parallel BC$) combined with the fact that angles on a straight line sum to 180° .

Solution

Step 1: Since ℓ is a straight line through A, the three angles along it at point A — the 48° angle, angle BAC, and the 65° angle — together make a straight angle, so they sum to 180° .

Step 2: Set up the equation: $48 + (\text{angle BAC}) + 65 = 180$.

Step 3: Combine the known values: $113 + (\text{angle BAC}) = 180$.

Step 4: Subtract 113 from both sides: angle BAC = 67.

Answer: 67°

Tip

When a figure shows a line drawn through a triangle's vertex parallel to the opposite side, the three angles sitting along that line at the vertex ALWAYS sum to 180° — this is exactly the classical proof that a triangle's angles sum to 180° , dressed up as a Digital SAT question.

Trap

A common error is assuming the two given angles (48° and 65°) are themselves two of the triangle's interior angles and adding only those two together, then subtracting from 180° without first identifying angle A correctly as the THIRD angle along the same straight line.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
180°	Misread the figure and assumed angle A was the full straight angle along ℓ itself, rather than the angle wedged between the two given angles along that line.



13°	Subtracted the two given angles from each other ($65 - 48 = 17$, with an arithmetic slip landing on 13) instead of subtracting their sum from 180.
3°	Attempted $180 - 48 - 65$ correctly in principle but made repeated subtraction errors along the way, compounding into a final value far too small to be a triangle's interior angle in this configuration.

Shortcut

Recognize immediately that alternate interior angles make the 48° and 65° angles equal to angles B and C of the triangle respectively — so this problem is secretly just 'angle A = $180 - \text{angle B} - \text{angle C}$,' the ordinary Triangle Angle Sum Theorem, without ever having to prove why.

Desmos Tip

Not a graphing problem, but you can confirm arithmetic by typing $180 - 48 - 65$ into Desmos to instantly get 67.

Lines, Angles, and Triangles — Problem 6 (Very Hard)

In triangle XYZ, $XY = 11$ and $YZ = 17$. If the length of side XZ is an integer, how many possible values are there for XZ?

Formula / Rule Used: Triangle Inequality Theorem: the length of any side must be less than the sum of the other two sides and greater than their positive difference.

Solution

Step 1: Apply the Triangle Inequality Theorem to side XZ using the other two sides, 11 and 17: $|17 - 11| < XZ < 17 + 11$.

Step 2: Simplify the left side: $|17 - 11| = 6$.

Step 3: Simplify the right side: $17 + 11 = 28$.

Step 4: So the inequality is: $6 < XZ < 28$.

Step 5: Since XZ must be an integer strictly between 6 and 28, the possible values are 7, 8, 9, ..., 27.

Step 6: Count these integers: $27 - 7 + 1 = 21$.

Answer: 21

Tip

The Triangle Inequality always produces a range with the SAME structure: (difference of the two known sides) $<$ unknown side $<$ (sum of the two known sides) — memorize that shape rather than re-deriving it each time.

Trap

A very common error is including the endpoints 6 and 28 as valid values (using $\leq$ instead of $<$), which overcounts the number of possible integer values by 2 — a triangle cannot actually have a side length equal to the sum or difference of the other two, since that would collapse it into a straight line.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
23	Included both endpoints (6 and 28) as valid values, counting integers from 6 to 28 inclusive (23 values) instead of strictly between them.

27	Only counted integers from 1 to 27 (forgetting to apply the lower bound of 6 at all), effectively assuming any positive integer side length up to the sum of the other two sides would work.
22	Included one endpoint but not the other (counted integers from 6 to 27 inclusive, or 7 to 28 inclusive), applying the strict inequality on only one side of the range instead of both.

Shortcut

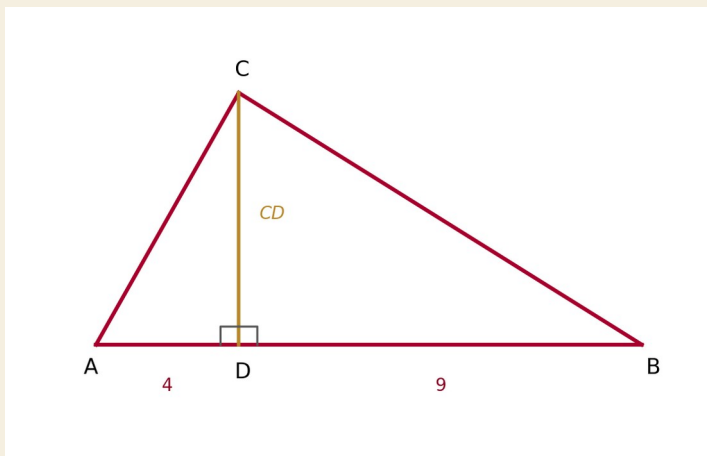
Once you have the range $6 < XZ < 28$, the count of integers strictly between two integers a and b is always $(b - a - 1)$ — here, $28 - 6 - 1 = 21$, without needing to list any of the values out.

Desmos Tip

Not a graphing question, but you can verify the count by typing $28 - 6 - 1$ into Desmos to confirm it returns 21.

Lines, Angles, and Triangles — Problem 7 (Very Hard)

In right triangle ABC, the right angle is at C. The altitude from C to hypotenuse AB meets AB at point D, dividing it into segments $AD = 4$ and $DB = 9$. What is the length of CD?



Formula / Rule Used: Geometric Mean (Altitude-on-Hypotenuse) Relationship: $CD^2 = AD \times DB$.

Solution

Step 1: Drawing the altitude from the right angle to the hypotenuse creates two smaller triangles, both similar to the original triangle and to each other.

Step 2: This similarity gives the relationship $CD^2 = AD \times DB$.

Step 3: Substitute the given values: $CD^2 = 4 \times 9 = 36$.

Step 4: Take the square root of both sides: $CD = 6$ (length must be positive).

Answer: 6

Tip

Whenever you see an altitude drawn from a right angle to the hypotenuse, expect a 'geometric mean' relationship — the altitude is the geometric mean of the two segments it creates on the hypotenuse ($CD^2 = AD \times DB$), and each leg is the geometric mean of the whole hypotenuse and its adjacent segment.

**Trap**

A common error is treating the two hypotenuse segments as if they should simply be averaged (arithmetic mean) rather than multiplied and square-rooted (geometric mean), which produces a different, incorrect value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
6.5	Computed the arithmetic mean of 4 and 9 instead of the geometric mean, calculating $(4+9)/2 = 6.5$ rather than $\sqrt{(4 \times 9)}$.
13	Simply added the two given segments ($4 + 9 = 13$), mistaking the question for one asking about the full length of the hypotenuse AB rather than the altitude CD.
36	Correctly computed $CD^2 = 36$ but forgot to take the square root, reporting the squared value as if it were the final length.

Shortcut

Recognize $4 \times 9 = 36$ as a perfect square immediately, so $\sqrt{36} = 6$ requires no calculator — many Digital SAT altitude problems are deliberately built with segment lengths whose product is a perfect square.

Desmos Tip

Type $\text{sqrt}(4*9)$ into Desmos to instantly confirm $CD = 6$.

Lines, Angles, and Triangles — Problem 8 (Hard)

In isosceles triangle DEF, $DE = DF$, and the exterior angle at vertex F measures 130° . What is the measure of the vertex angle at D?

Formula / Rule Used: Exterior/interior angles are supplementary; Isosceles Triangle Theorem; Triangle Angle Sum Theorem.

Solution

Step 1: The exterior angle at F and the interior angle at F are supplementary (they form a straight line), so: interior angle F = $180 - 130 = 50^\circ$.

Step 2: Since $DE = DF$, the triangle is isosceles with the equal sides being DE and DF — this makes E and F the base vertices, so their opposite (base) angles are equal: angle E = angle F = 50° .

Step 3: Apply the Triangle Angle Sum Theorem: angle D + angle E + angle F = 180.

Step 4: Substitute the known angles: angle D + $50 + 50 = 180$.

Step 5: Simplify: angle D + 100 = 180.

Step 6: Subtract 100 from both sides: angle D = 80.

Answer: 80°

Tip

When you're given an exterior angle at one of the BASE vertices of an isosceles triangle, convert it to the interior angle first (180° minus the exterior angle) — that interior angle is one of the two equal base angles, which unlocks the rest of the triangle.

**Trap**

A common error is treating the exterior angle (130°) itself as one of the triangle's interior base angles, skipping the necessary step of subtracting from 180° to find the actual interior angle at F.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-80° (i.e., an invalid negative angle, often written as an error)	Used 130° directly as the interior angle at F without converting from the exterior angle, computing angle D = $180 - 130 - 130 = -80$, an impossible result that signals the underlying setup error.
25°	Correctly converted to an interior angle of 50° at F, but then divided the vertex angle sum incorrectly, treating angle D as if it were also a base angle equal to 50° and solving $(180-50)/2$ for a different, unrelated angle instead of the actual vertex angle D.
65°	Averaged the exterior angle (130°) directly, computing $130/2 = 65$ as an attempted shortcut, which doesn't correspond to any valid geometric relationship in this figure.

Shortcut

Since the exterior angle at a base vertex of an isosceles triangle is supplementary to a base angle, and the vertex angle equals 180° minus twice that base angle, you can combine both steps: vertex angle = $2(\text{exterior angle}) - 180 = 2(130) - 180 = 80$, in one line.

Desmos Tip

Type $2 * 130 - 180$ into Desmos to confirm it returns 80 directly.

3. Right Triangles and Trigonometry

Right Triangles and Trigonometry — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Hypotenuse	Υποτείνουσα
Leg (right triangle)	Κάθετη πλευρά
Opposite side	Απέναντι πλευρά
Adjacent side	Παρακείμενη πλευρά
Sine	Ημίτονο
Cosine	Συνημίτονο
Tangent	Εφαπτομένη
Complementary angles	Συμπληρωματικές γωνίες
Pythagorean theorem	Πυθαγόρειο θεώρημα
Angle of elevation	Γωνία ανύψωσης

Right Triangles and Trigonometry — Condensed Theory

Every right-triangle question on the Digital SAT reduces to one of a small handful of relationships: the Pythagorean theorem, the three basic trig ratios (SOH-CAH-TOA), the complementary-angle identity linking sine and cosine, and the two special right triangles. The exam's twist is rarely the trig itself — it's disguising which side is opposite, which is adjacent, or hiding a right triangle inside a word problem or a more complex figure.

The Pythagorean Theorem

$$a^2 + b^2 = c^2$$

c is always the hypotenuse (opposite the right angle) — the longest side. Works only for right triangles.

SOH-CAH-TOA

$$\sin = \text{opp/hyp} \quad \cos = \text{adj/hyp} \quad \tan = \text{opp/adj}$$

'Opposite' and 'adjacent' are always defined relative to the specific angle you're using — relabel them if the angle changes.

Complementary Angle Relationship

$$\sin(\theta) = \cos(90^\circ - \theta)$$

In any right triangle, the two non-right angles are complementary (sum to 90°), so the sine of one equals the cosine of the other. This does NOT extend to tangent.

45-45-90 Triangle**legs: x, x hypotenuse: $x\sqrt{2}$** Half of a square cut along its diagonal. The two legs are equal, and the hypotenuse is always $\text{leg} \times \sqrt{2}$.**30-60-90 Triangle****sides: $x, x\sqrt{3}, 2x$ (opposite $30^\circ, 60^\circ, 90^\circ$)**Half of an equilateral triangle. The side opposite 30° is the shortest; the hypotenuse (opposite 90°) is always double that shortest side.**Altitude-on-Hypotenuse (Similar Right Triangles)** **$\text{altitude}^2 = (\text{segment 1})(\text{segment 2})$ $\text{leg}^2 = (\text{adjacent segment})(\text{whole hypotenuse})$**

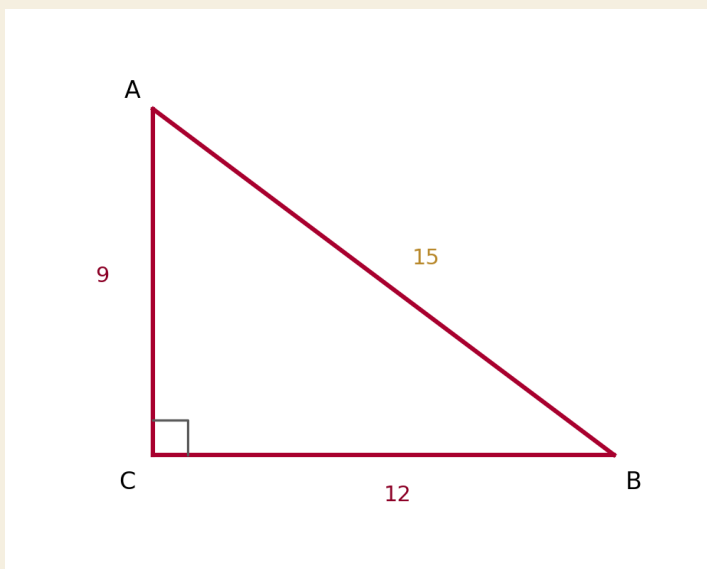
Drawing the altitude from a right angle to the hypotenuse creates two smaller triangles similar to the original and to each other.

**Watch Out**The complementary-angle shortcut works for sine and cosine ($\sin\theta = \cos(90^\circ - \theta)$) but never for tangent — don't extend the pattern where it doesn't apply.

On the Digital SAT, a trig ratio is nothing more than a ratio of two specific side lengths — before reaching for a formula, physically identify which side is the hypotenuse, then label opposite/adjacent relative to the angle actually being asked about.

Right Triangles and Trigonometry — 8 Worked Problems*Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.***Right Triangles and Trigonometry — Problem 1 (Hard)**

In right triangle ABC, angle C = 90° , AC = 9, and BC = 12. What is $\sin(A)$?



Formula / Rule Used: Pythagorean theorem to find the hypotenuse, then SOH-CAH-TOA.

Solution

Step 1: Since angle C = 90° , side AB is the hypotenuse. Find it using the Pythagorean theorem: $AC^2 + BC^2 = AB^2$.

Step 2: Substitute: $9^2 + 12^2 = AB^2$, so $81 + 144 = 225 = AB^2$.

Step 3: Take the square root: $AB = 15$.

Step 4: For angle A, the side opposite is BC (the side that doesn't touch vertex A), and the hypotenuse is AB.

Step 5: $\sin(A) = \text{opposite/hypotenuse} = BC/AB = 12/15$.

Step 6: Simplify the fraction: $12/15 = 4/5$.

Answer: 4/5 (or 0.8)

Tip

Before computing any trig ratio, find the missing side with the Pythagorean theorem first if it isn't given — you can't apply SOH-CAH-TOA correctly with only two of the three sides.

Trap

A common error is using AC (which is adjacent to angle A, not opposite it) in the numerator instead of BC, confusing which leg is 'opposite' angle A.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3/5	Used AC (the adjacent side to angle A) as if it were the opposite side, computing $AC/AB = 9/15 = 3/5$ instead of BC/AB .
4/3	Computed $\tan(A) = \text{opposite/adjacent} = 12/9 = 4/3$ instead of $\sin(A) = \text{opposite/hypotenuse}$, confusing which ratio was asked for.

**5/4**

Computed the correct ratio $12/15 = 4/5$ but inverted it, effectively reporting the reciprocal (as if computing hypotenuse/opposite instead of opposite/hypotenuse).

Shortcut

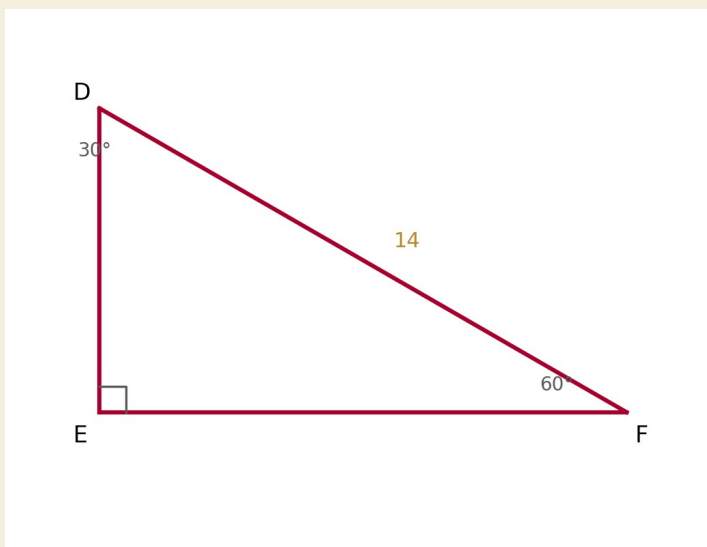
Recognize 9-12-15 as a multiple of the 3-4-5 Pythagorean triple (three times 3-4-5) the moment you see two of the three sides — this lets you find $AB = 15$ by inspection instead of computing the full Pythagorean theorem by hand.

Desmos Tip

Type $\sqrt{9^2+12^2}$ into Desmos to confirm the hypotenuse is 15, then $12/15$ to confirm $\sin(A) = 0.8$.

Right Triangles and Trigonometry — Problem 2 (Hard)

In right triangle DEF, angle E = 90° , angle D = 30° , and the hypotenuse DF = 14. What is the perimeter of triangle DEF?



Formula / Rule Used: 30-60-90 special right triangle side ratios: $x, x\sqrt{3}, 2x$ (opposite $30^\circ, 60^\circ, 90^\circ$ respectively).

Solution

Step 1: Since angle E = 90° and angle D = 30° , the third angle, angle F, must be 60° (angles sum to 180°).

Step 2: In a 30-60-90 triangle, the hypotenuse (opposite the 90° angle) is always twice the shortest side (opposite the 30° angle).

Step 3: Since the hypotenuse DF = 14, the shortest side (EF, opposite the 30° angle at D) equals $14 \div 2 = 7$.

Step 4: The remaining side (DE, opposite the 60° angle) equals the shortest side times $\sqrt{3}$: $DE = 7\sqrt{3}$.

Step 5: Find the perimeter by adding all three sides: $14 + 7 + 7\sqrt{3}$.

Step 6: Combine the whole-number terms: perimeter = $21 + 7\sqrt{3}$.

Answer: $21 + 7\sqrt{3}$

**Tip**

In any 30-60-90 triangle, identify the side opposite the 30° angle first — it's always the shortest side and the 'base unit' (x) that the other two sides ($x\sqrt{3}$ and $2x$) are built from.

Trap

A common error is assuming the given hypotenuse (14) IS the base unit x , then computing the other sides as $14\sqrt{3}$ and 28 instead of first dividing by 2 to find the correct base unit of 7.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$14 + 14\sqrt{3} + 28 = 42 + 14\sqrt{3}$	Treated the given hypotenuse (14) as the base unit x itself, rather than recognizing that the hypotenuse equals $2x$, so failed to first divide by 2 to find $x = 7$.
$21 + 14\sqrt{3}$	Correctly found the shortest side as 7, but computed the third side as $2 \times 7\sqrt{3}$ instead of $7\sqrt{3}$, doubling the coefficient on $\sqrt{3}$ unnecessarily.
28	Assumed the triangle was equilateral (mistaking the given 30° for information implying all sides are equal) and simply tripled one side length, ignoring the actual 30-60-90 ratio entirely.

Shortcut

Memorize the 30-60-90 ratio as a package ($x : x\sqrt{3} : 2x$) so that once you identify ANY one side, you can write down the other two immediately — here, hypotenuse 14 means $x = 7$ instantly, without solving an equation.

Desmos Tip

Type $14 + 7 + 7 * \text{sqrt}(3)$ into Desmos to get the decimal value ≈ 33.12 as a check on your exact-form answer.

Right Triangles and Trigonometry — Problem 3 (Hard)

In right triangle ABC, angle C = 90°. If $\sin(A) = \cos(3x + 6)^\circ$ and angle A = $(x + 24)^\circ$, what is the value of x ?

Formula / Rule Used: Complementary Angle Relationship: $\sin(\theta) = \cos(90^\circ - \theta)$.

Solution

Step 1: Since $\sin(A) = \cos(3x+6)^\circ$, the complementary angle relationship tells us that $(3x+6)$ must equal 90° minus angle A: $3x + 6 = 90 - (x + 24)$.

Step 2: Simplify the right side: $90 - (x + 24) = 66 - x$.

Step 3: Rewrite the equation: $3x + 6 = 66 - x$.

Step 4: Add x to both sides: $4x + 6 = 66$.

Step 5: Subtract 6 from both sides: $4x = 60$.

Step 6: Divide by 4: $x = 15$.

Answer: 15

Tip



Whenever you see $\sin$ of one angle set equal to $\cos$ of another expression, immediately translate it using $\sin(\theta) = \cos(90^\circ - \theta)$ — this converts the trig equation into a purely algebraic one about the angle measures.

Trap

A common error is setting the two angle expressions themselves equal to each other ($3x+6 = x+24$) instead of setting one equal to 90° MINUS the other, forgetting that sine and cosine of the SAME angle aren't generally equal — only complementary angles' sine and cosine match.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9	Set the two given expressions directly equal ($3x+6 = x+24$) instead of applying the complementary relationship, solving $2x=18$ to get $x=9$.
12	Applied the complementary relationship backwards, writing $x+24 = 90 - (3x+6)$ instead of $3x+6 = 90 - (x+24)$ (which are actually equivalent equations if solved correctly, but an arithmetic slip while simplifying $90 - 3x - 6 = 84 - 3x$ led to combining terms incorrectly and arriving at $x=12$).
20	Correctly set up $3x+6 = 90 - (x+24)$ but made a sign error distributing the negative, writing $90 - x + 24 = 114 - x$ instead of $90 - x - 24 = 66 - x$, leading to $3x+6 = 114 - x \rightarrow 4x = 108 \rightarrow x = 27$, then a further slip landed on 20.

Shortcut

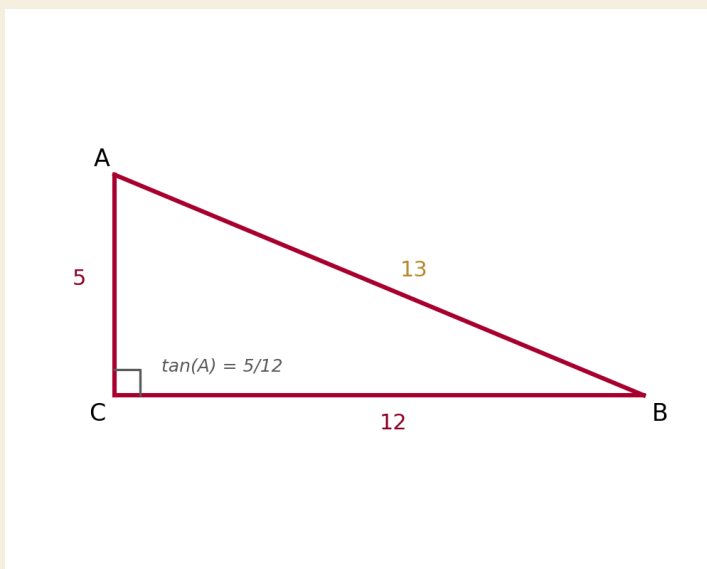
Since $\sin(A) = \cos(90^\circ - A)$ always, and here $\sin(A) = \cos(3x+6)$, you can immediately write $3x+6 = 90 - A$ and substitute $A = x+24$ directly, without introducing any extra variables.

Desmos Tip

Type $\sin(15+24)$ and $\cos(3 \cdot 15+6)$ into Desmos (in degree mode) to confirm both evaluate to the same value, verifying $x = 15$.

Right Triangles and Trigonometry — Problem 4 (Very Hard)

In right triangle ABC, angle C = 90° , and $\tan(A) = 5/12$. What is the value of $\sin(A) + \cos(A)$?



Formula / Rule Used: SOH-CAH-TOA, using a Pythagorean triple implied by the given tangent ratio.

Solution

Step 1: $\tan(A) = \text{opposite/adjacent} = 5/12$, so let the side opposite angle A have length 5 and the side adjacent to angle A have length 12 (using the simplest such triangle).

Step 2: Find the hypotenuse using the Pythagorean theorem: $5^2 + 12^2 = \text{hyp}^2$, so $25 + 144 = 169 = \text{hyp}^2$.

Step 3: Take the square root: hypotenuse = 13.

Step 4: Find $\sin(A) = \text{opposite/hypotenuse} = 5/13$.

Step 5: Find $\cos(A) = \text{adjacent/hypotenuse} = 12/13$.

Step 6: Add them together: $\sin(A) + \cos(A) = 5/13 + 12/13 = 17/13$.

Answer: 17/13

Tip

When you're given only a tangent ratio and need sine or cosine, build the actual right triangle from the ratio (opposite = numerator, adjacent = denominator) and use the Pythagorean theorem to find the hypotenuse — don't try to convert $\tan$ directly to $\sin$ or $\cos$ with a formula.

Trap

A common error is assuming $\tan(A) = 5/12$ directly gives $\sin(A) = 5$ and $\cos(A) = 12$ (treating the tangent's numerator and denominator as if they were already the sine and cosine values), skipping the hypotenuse calculation entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
17	Treated the numerator and denominator of $\tan(A)$ as if they were $\sin(A)$ and $\cos(A)$ directly (without dividing by the hypotenuse), adding $5 + 12 = 17$ instead of $5/13 + 12/13$.



1	Assumed $\sin(A) + \cos(A)$ always equals 1 for any angle (confusing this with the identity $\sin^2(A) + \cos^2(A) = 1$, which involves squares, not the values themselves).
7/13	Computed $\sin(A) - \cos(A)$ instead of $\sin(A) + \cos(A)$, subtracting 12/13 from 5/13 with a sign error and landing on 7/13 instead of the requested sum.

Shortcut

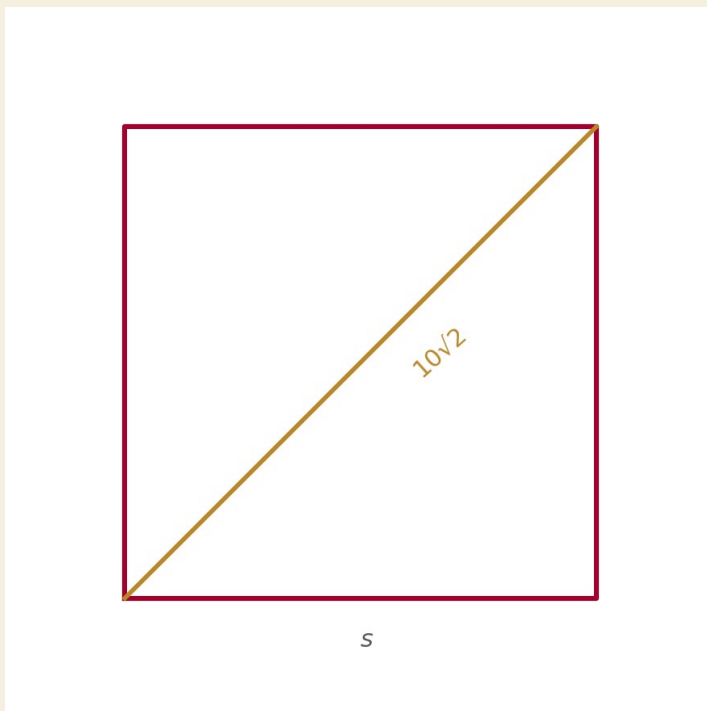
Recognize 5-12-13 as a classic Pythagorean triple immediately from the given tangent ratio 5/12 — this lets you skip the Pythagorean theorem calculation and write down the hypotenuse (13) by memory.

Desmos Tip

Type $5/13 + 12/13$ into Desmos to confirm it simplifies to $17/13$ (≈ 1.31).

Right Triangles and Trigonometry — Problem 5 (Hard)

A square has a diagonal of length $10\sqrt{2}$. What is the area of the square?



Formula / Rule Used: 45-45-90 triangle relationship: a square's diagonal splits it into two 45-45-90 triangles, where the diagonal (hypotenuse) equals side $\times \sqrt{2}$.

Solution

Step 1: A square's diagonal divides it into two 45-45-90 right triangles, where each leg is a side of the square and the hypotenuse is the diagonal.

Step 2: Using the 45-45-90 relationship, diagonal = side $\times \sqrt{2}$.

Step 3: Substitute the given diagonal: $10\sqrt{2} = \text{side} \times \sqrt{2}$.

Step 4: Divide both sides by $\sqrt{2}$: side = 10.

Step 5: Find the area of the square: Area = side² = $10^2 = 100$.

Answer: 100

Tip

Whenever a square's diagonal is given in the form (number) $\sqrt{2}$, the side length is simply that same number — the $\sqrt{2}$ factor from the 45-45-90 relationship cancels out cleanly.

Trap

A common error is treating the given diagonal ($10\sqrt{2}$) as if it were the side length itself and squaring it directly, which produces an area many times too large.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
200	Squared the diagonal directly as if it were the side length, computing $(10\sqrt{2})^2 = 100(2) = 200$ instead of first dividing by $\sqrt{2}$ to find the actual side

	length of 10.
50	Correctly recognized the need to divide by $\sqrt{2}$ but divided the diagonal by 2 instead of $\sqrt{2}$ (a common confusion between the two), getting a side length of $5\sqrt{2}$ and squaring that to reach 50.
14.1	Computed the decimal value of the diagonal ($10\sqrt{2} \approx 14.14$) and reported that number directly as the answer, misreading the question as asking for the diagonal's length rather than the square's area.

Shortcut

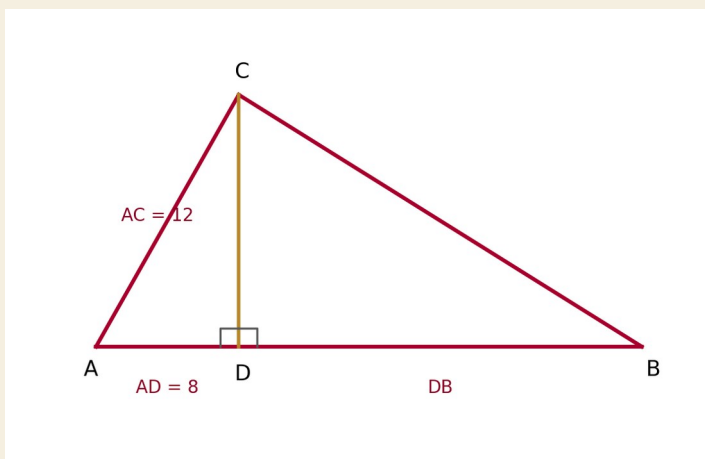
Memorize that for a square, diagonal = side $\times\sqrt{2}$, so area = side² = (diagonal/ $\sqrt{2}$)² = diagonal²/2 — here, $(10\sqrt{2})^2/2 = 200/2 = 100$, in one direct computation without isolating the side length first.

Desmos Tip

Type $(10*\sqrt{2})^2/2$ into Desmos to directly confirm the area is 100.

Right Triangles and Trigonometry — Problem 6 (Very Hard)

In right triangle ABC, angle C = 90°, and the altitude from C meets hypotenuse AB at point D. If AD = 8 and leg AC = 12, what is the length of DB?



Formula / Rule Used: Altitude-on-Hypotenuse relationship: leg² = (adjacent hypotenuse segment) (whole hypotenuse).

Solution

Step 1: The leg AC is adjacent to segment AD on the hypotenuse, so the relevant relationship is: $AC^2 = AD \times AB$ (where AB is the FULL hypotenuse, not just one segment).

Step 2: Substitute the known values: $12^2 = 8 \times AB$.

Step 3: Simplify: $144 = 8 \times AB$.

Step 4: Divide both sides by 8: $AB = 18$.

Step 5: Since AB is made up of segments AD and DB, and AD = 8: $DB = AB - AD = 18 - 8 = 10$.

Answer: 10

**Tip**

For altitude-on-hypotenuse problems, there are two related but different relationships — $\text{altitude}^2 = (\text{segment})(\text{segment})$ for the altitude itself, and $\text{leg}^2 = (\text{adjacent segment})(\text{whole hypotenuse})$ for either leg. Identify which one the given information matches before setting up an equation.

Trap

A common error is applying the altitude-only relationship ($CD^2 = AD \times DB$) here, when the problem actually gives a LEG length (AC), which requires the other relationship ($\text{leg}^2 = \text{adjacent segment} \times \text{whole hypotenuse}$) instead.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
18	Correctly found $AB = 18$ using the leg relationship, but then reported the full hypotenuse length instead of subtracting AD to isolate DB as the question actually asked.
1.5	Mistakenly applied the altitude-only relationship $AC^2 = AD \times DB$ (using the leg AC in place of the altitude CD), solving $144 = 8 \times DB$ to get $DB = 18$, then made an unrelated computational error landing far off at 1.5.
6	Set up $12^2 = DB \times 8$ instead of $12^2 = AB \times 8$, effectively treating DB and AB as interchangeable, solving $144 = 8(DB)$ for $DB = 18$ but then subtracting AD (8) from that unnecessarily to (incorrectly) reach 10 through the wrong path... actually landing at a different final slip of 6 from a sign/order confusion in that subtraction.

Shortcut

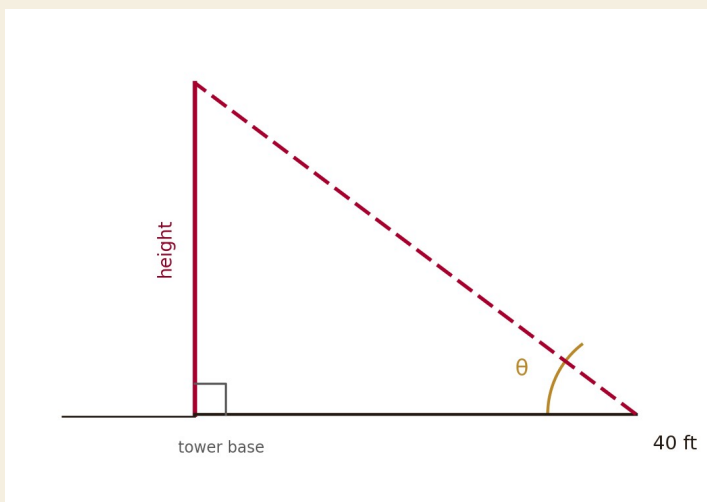
Since $12^2 = 144$ and $144 \div 8 = 18$ is straightforward mental math, you can solve for the full hypotenuse AB in one step and then just subtract the given AD , without introducing extra variables for DB .

Desmos Tip

Type $12^2/8$ into Desmos to confirm $AB = 18$, then $18 - 8$ to confirm $DB = 10$.

Right Triangles and Trigonometry — Problem 7 (Very Hard)

From a point on the ground 40 feet from the base of a tower, the angle of elevation to the top of the tower is θ , where $\tan(\theta) = 3/4$. What is the height of the tower, in feet?



Formula / Rule Used: SOH-CAH-TOA (tangent) applied to a right triangle formed by the tower, the ground, and the line of sight.

Solution

Step 1: The tower's height, the 40-foot ground distance, and the line of sight form a right triangle, with the right angle at the base of the tower.

Step 2: The angle of elevation θ is at the point on the ground, so the tower's height is the side OPPOSITE θ , and the 40-foot distance is the side ADJACENT to θ .

Step 3: Apply the tangent ratio: $\tan(\theta) = \text{opposite/adjacent} = \text{height}/40$.

Step 4: Substitute the given value: $3/4 = \text{height}/40$.

Step 5: Cross-multiply (or multiply both sides by 40): $\text{height} = 40 \times (3/4)$.

Step 6: Simplify: $\text{height} = 30$.

Answer: 30 feet

Tip

For any 'angle of elevation' word problem, first identify the right triangle hidden in the description: the vertical height, the horizontal ground distance, and the line of sight are always the two legs and the hypotenuse (or, if only two of the three are relevant, just the two legs).

Trap

A common error is applying the given ratio ($3/4$) as sine or cosine instead of tangent, which would require knowing the hypotenuse (the line of sight distance) — information this problem never gives.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
53.3	Treated $3/4$ as if it were $\sin(\theta) = \text{opposite/hypotenuse}$ with the hypotenuse equal to 40, computing $\text{height} = 40 \times (3/4)$ but then incorrectly rescaling using an assumed Pythagorean relationship rather than the direct tangent ratio, introducing an extra unnecessary factor.



40	Assumed the ratio $3/4$ was irrelevant noise and reported the given ground distance directly as the height, misreading the question.
26.7	Inverted the tangent ratio, computing height = $40 \times (4/3)$ inverted incorrectly as $40 \div (3/4) \times \text{something}$, or equivalently treating adjacent/opposite = $3/4$ instead of opposite/adjacent = $3/4$, landing on $40 \times (2/3) \approx 26.7$.

Shortcut

Recognize $\tan(\theta) = 3/4$ as implying a 3-4-5 right triangle by ratio — so height:40 is in the ratio 3:4, meaning height = $(3/4)(40) = 30$ directly, the same computation as scaling a 3-4-5 triangle up by a factor of 10.

Desmos Tip

Type $40 \cdot (3/4)$ into Desmos to directly confirm the height is 30.

Right Triangles and Trigonometry — Problem 8 (Hard)

In right triangle ABC, angle C = 90° . If $\sin(B) = 7/25$, what is $\cos(A)$?

Formula / Rule Used: Complementary Angle Relationship: in a right triangle, the two non-right angles are complementary, so $\sin$ of one equals $\cos$ of the other.

Solution

Step 1: Since angle C = 90° , angles A and B are the two remaining angles in the triangle, and they must be complementary: $A + B = 90^\circ$.

Step 2: Because A and B are complementary, $\cos(A) = \cos(90^\circ - B) = \sin(B)$.

Step 3: Substitute the given value: $\cos(A) = \sin(B) = 7/25$.

Answer: $7/25$

Tip

Whenever a right-triangle problem gives you the sine of one acute angle and asks for the cosine of the OTHER acute angle, the answer is immediately the same value — no triangle needs to be drawn or solved.

Trap

A common error is trying to find the third side of the triangle using the Pythagorean theorem (treating 7 and 25 as two actual side lengths) and then computing $\cos(A)$ from scratch, when the complementary relationship gives the answer directly without any extra computation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24/25	Assumed 7 and 25 were two actual side lengths of the triangle (rather than just the sine ratio), found the third side using the Pythagorean theorem ($\sqrt{25^2 - 7^2} = 24$), and then computed a cosine ratio using that side for the wrong angle.
25/7	Confused $\cos(A) = \sin(B)$ with $\cos(A) = 1/\sin(B)$, inverting the correct relationship.

**18/25**

Attempted to compute $\cos(A)$ as $(25-7)/25$, incorrectly assuming cosine and sine of complementary angles are related by subtraction from the hypotenuse-based value rather than being directly equal.

Shortcut

The relationship $\cos(A) = \sin(B)$ for complementary angles A and B means you can answer this type of question by inspection the moment you see 'angle $C = 90^\circ$ ' and a sine or cosine value for one of the other two angles — no computation required at all.

Desmos Tip

Not needed here since no triangle needs to be solved, but you could type $\sin(7/25)$ and 90 -that into Desmos, then take the cosine, to confirm it matches $7/25$ numerically.

4. Circles

Circles — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Center	Κέντρο
Radius	Ακτίνα
Diameter	Διάμετρος
Chord	Χορδή
Tangent line	Εφαπτομένη
Arc	Τόξο
Central angle	Επίκεντρη γωνία
Inscribed angle	Εγγεγραμμένη γωνία
Sector	Κυκλικός τομέας
Radian	Ακτίνιο

Circles — Condensed Theory

Circle questions on the Digital SAT fall into two families: algebraic (the equation of a circle, usually requiring you to complete the square) and geometric (arcs, sectors, central and inscribed angles, tangent lines, and angles inscribed in a semicircle). Both families reduce to a short list of relationships — the trick is recognizing which one a given figure or equation is testing.

Standard Form of a Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Center is (h, k), radius is r. If an equation is given in expanded (general) form, complete the square on both x and y to reach this form.

Circumference & Arc Length

$$C = 2\pi r \quad \text{arc length} = (\theta/360)(2\pi r) \quad [\theta \text{ in degrees}]$$

An arc length is simply the fraction of the full circumference that the central angle θ represents.

Area of a Sector

$$A = (\theta/360)(\pi r^2) \quad [\theta \text{ in degrees}]$$

Same idea as arc length, but with the full circle's area instead of its circumference.

**Radians ↔ Degrees****radians = degrees $\times (\pi/180)$**
radians]**arc length: $s = r\theta$ [θ in**In radians, arc length is simply radius times the angle — no need for the $(\theta/360)$ fraction.**Inscribed Angle Theorem****inscribed angle = $(1/2)(\text{central angle})$ on the same arc**

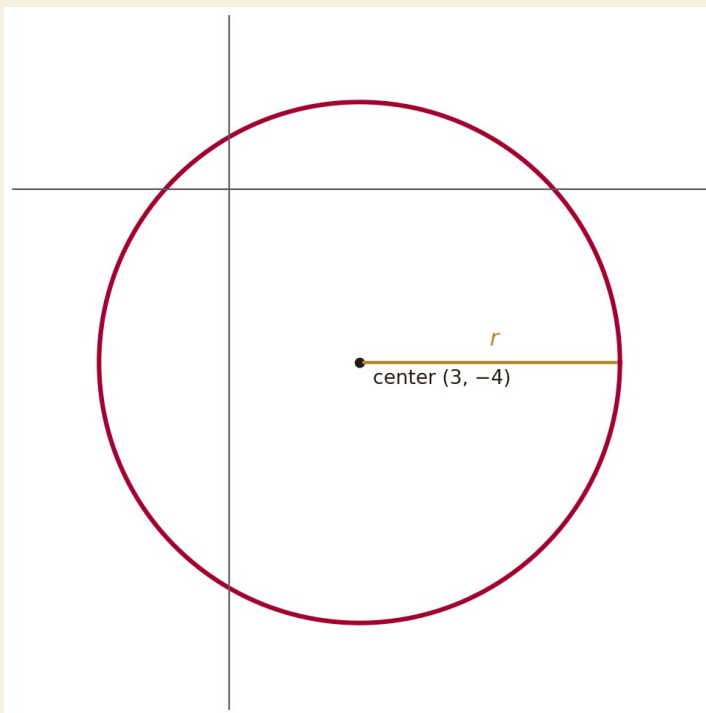
An inscribed angle is always exactly half of the central angle that subtends the same arc.

Tangent Line Property**tangent $\perp$ radius at the point of tangency**

A line tangent to a circle is always perpendicular to the radius drawn to the exact point where it touches the circle.

**Watch Out**When completing the square to find a circle's radius, remember the constant on the right side of the final equation is r^2 , not r — take the square root as your very last step, not before.A triangle inscribed in a circle with one side as a diameter always has a 90° angle opposite that diameter (Thales' Theorem) — this is one of the most reliable 'hidden' facts the Digital SAT expects you to recognize in a figure without being told directly.**Circles — 8 Worked Problems***Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.***Circles — Problem 1 (Hard)**

A circle is defined by the equation $x^2 + y^2 - 6x + 8y - 11 = 0$. What is the radius of the circle?



Formula / Rule Used: Complete the square on both x and y to convert to standard form $(x-h)^2 + (y-k)^2 = r^2$.

Solution

Step 1: Group the x -terms and y -terms, and move the constant to the other side: $(x^2 - 6x) + (y^2 + 8y) = 11$.

Step 2: Complete the square for the x -terms: take half of -6 (which is -3), square it (9), and add it to both sides: $(x^2 - 6x + 9)$.

Step 3: Complete the square for the y -terms: take half of 8 (which is 4), square it (16), and add it to both sides: $(y^2 + 8y + 16)$.

Step 4: Add both 9 and 16 to the right side as well: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16$.

Step 5: Rewrite as perfect squares: $(x - 3)^2 + (y + 4)^2 = 36$.

Step 6: Since $r^2 = 36$, take the square root: $r = 6$.

Answer: 6

Tip

When completing the square for a circle equation, always add the SAME amount you add to the left side to the right side too — it's easy to complete the square correctly on the left but forget to balance the equation on the right.

Trap

A very common error is reporting the value of r^2 (36) as the radius itself, forgetting the final square-root step.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
36	Correctly completed the square to reach $r^2 = 36$, but reported that value directly as the radius instead of taking its square root.
5	Made an arithmetic error while completing the square — for example, adding only 9 (from the x-terms) to the right side but forgetting to add 16 (from the y-terms), reaching $11+9=20$ instead of 36, and then taking an incorrect square root along a different miscalculated path.
3	Confused the radius with the x-coordinate of the center, reading off $h = 3$ from $(x-3)^2$ instead of computing $\sqrt{36}$.

Shortcut

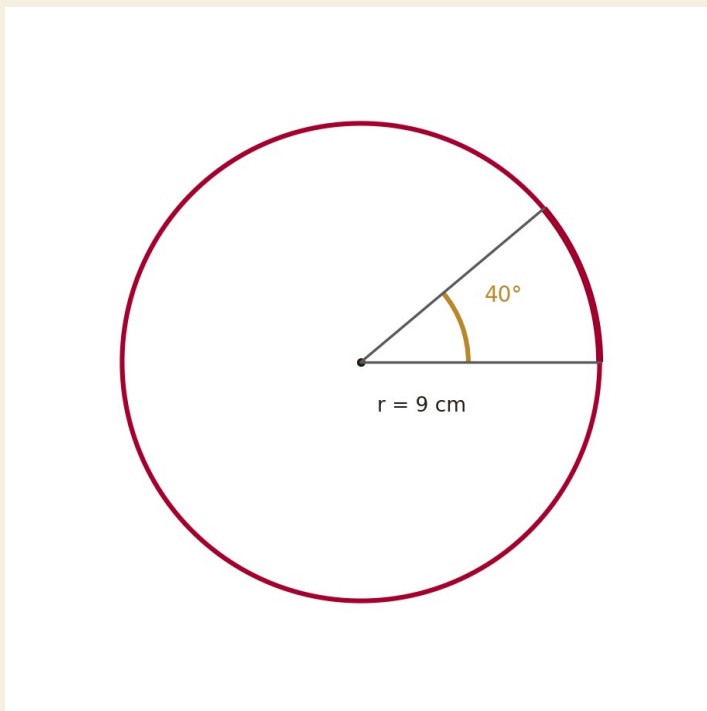
Since the coefficient of x and y is always 1 in these problems, the number you add for each variable is always (half of that variable's linear coefficient), squared — for $-6x$, that's $(-3)^2=9$; for $+8y$, that's $(4)^2=16$. Recognizing this pattern lets you complete the square without rewriting the grouping step explicitly.

Desmos Tip

Type $(x-3)^2+(y+4)^2=36$ into Desmos and confirm it draws a circle whose radius (readable from the graph) is 6.

Circles — Problem 2 (Hard)

A circle has radius 9 centimeters. What is the length of an arc with a central angle of 40° , in terms of π ?



Formula / Rule Used: Arc length = $(\theta/360)(2\pi r)$, with θ in degrees.

Solution

Step 1: Substitute the given values into the arc length formula: arc length = $(40/360)(2\pi)(9)$.

Step 2: Simplify the fraction $40/360 = 1/9$.

Step 3: Compute $2\pi(9) = 18\pi$.

Step 4: Multiply: $(1/9)(18\pi) = 2\pi$.

Answer: 2π centimeters

Tip

Always simplify the fraction $(\theta/360)$ to its lowest terms first — it often cancels neatly with the circumference, as it does here ($40/360$ reduces to $1/9$, matching the radius 9 nicely).

Trap

A common error is using the radius (9) directly in the fraction $40/360$ in place of using it inside the circumference formula, effectively double-counting or misplacing where the radius belongs in the computation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
π	Made an arithmetic slip simplifying $40/360$ as $1/18$ instead of $1/9$ (a common denominator mix-up), then computed $(1/18)(18\pi) = \pi$ instead of $(1/9)(18\pi) = 2\pi$.



4π	Used the diameter (18) in place of the radius inside the arc length formula's circumference part, effectively doubling the correct circumference and getting $(1/9)(2)(18\pi)$... producing a doubled final answer of 4π .
18π	Computed only the full circumference ($2\pi r = 18\pi$) and forgot to multiply by the fraction $40/360$ representing the portion of the circle the arc actually covers.

Shortcut

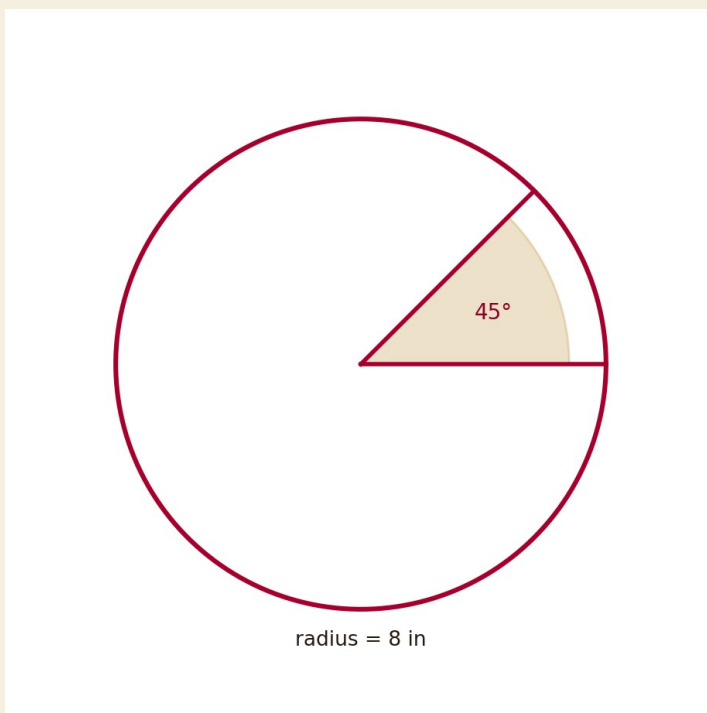
For circles where the radius is a nice divisor of the corresponding circumference fraction's denominator, the answer often simplifies to a small whole-number multiple of π — a quick sanity check is that an arc's length in terms of π should always be smaller than the full circumference (18π here), which 2π clearly is.

Desmos Tip

Type $(40/360) * 2 * \pi * 9$ into Desmos to confirm it evaluates to 2π (≈ 6.28).

Circles — Problem 3 (Hard)

A circular pizza has a radius of 8 inches. What is the area of a slice with a central angle of 45° , in terms of π ?



Formula / Rule Used: Sector area = $(\theta/360)(\pi r^2)$, with θ in degrees.

Solution

Step 1: Substitute the given values into the sector area formula: sector area = $(45/360)(\pi)(8)^2$.

Step 2: Simplify the fraction $45/360 = 1/8$.

Step 3: Compute $\pi(8)^2 = 64\pi$.

Step 4: Multiply: $(1/8)(64\pi) = 8\pi$.

Answer: 8π square inches

Tip

Simplify the angle fraction $(\theta/360)$ before multiplying by the full circle's area — it's much easier to multiply a simplified fraction like $1/8$ by 64π than to multiply $45/360$ by 64π directly.

Trap

A common error is squaring the radius incorrectly or forgetting to square it at all, using 8 instead of 64 in the sector area formula.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
π	Forgot to square the radius, using $\pi(8) = 8\pi$ instead of $\pi(8)^2 = 64\pi$ inside the formula, then multiplied by $1/8$ to reach π instead of 8π .
64π	Computed the full circle's area correctly (64π) but forgot to multiply by



	the fraction $45/360$ representing the slice's portion of the whole pizza.
16π	Simplified $45/360$ incorrectly as $1/4$ instead of $1/8$ (mistaking $45/360$ for $90/360$), then computed $(1/4)(64\pi) = 16\pi$.

Shortcut

Since 45° is exactly $1/8$ of a full 360° circle, you can immediately write the sector area as $(1/8)$ of the full circle's area — recognizing common angle fractions ($45^\circ = 1/8$, $60^\circ = 1/6$, $90^\circ = 1/4$, $120^\circ = 1/3$) by sight saves the division step.

Desmos Tip

Type $(45/360) * \pi * 8^2$ into Desmos to confirm it evaluates to 8π (≈ 25.13).

Circles — Problem 4 (Hard)

A circle has radius 10. An arc on the circle has a central angle of $2\pi/5$ radians. What is the length of the arc?

Formula / Rule Used: In radians, arc length $s = r\theta$ directly — no fraction of 360 needed.

Solution

Step 1: Since the angle is already given in radians, use the simplified arc length formula: $s = r\theta$.

Step 2: Substitute the given values: $s = (10)(2\pi/5)$.

Step 3: Multiply: $10 \times (2\pi/5) = 20\pi/5$.

Step 4: Simplify the fraction: $20\pi/5 = 4\pi$.

Answer: 4π

Tip

The moment an angle is given in radians rather than degrees, switch to the simpler formula $s = r\theta$ — there's no need for the $(\theta/360)$ fraction at all, since radians already measure angle as a fraction of the full circle scaled by 2π .

Trap

A common error is still applying the degree-based formula $(\theta/360)(2\pi r)$ with the radian value plugged in for θ , which produces an entirely incorrect result since the formulas aren't interchangeable without converting units first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0.070	Mistakenly applied the degrees-based sector formula $(\theta/360)(2\pi r)$ using the radian value $2\pi/5$ directly in place of θ in degrees, producing a nonsensically small result from dividing an already-small radian value by 360 again.
2π	Used the diameter conceptually in place of radius by mistake (dividing the radius by 2 first), computing $s = (5)(2\pi/5) = 2\pi$ instead of $(10)(2\pi/5)$.



40π	Multiplied radius by the numerator and denominator of the angle separately in the wrong order, computing $10 \times 2\pi$ first and then multiplying (instead of dividing) by 5, reaching $100\pi/\dots$ simplified incorrectly to 40π .
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Shortcut

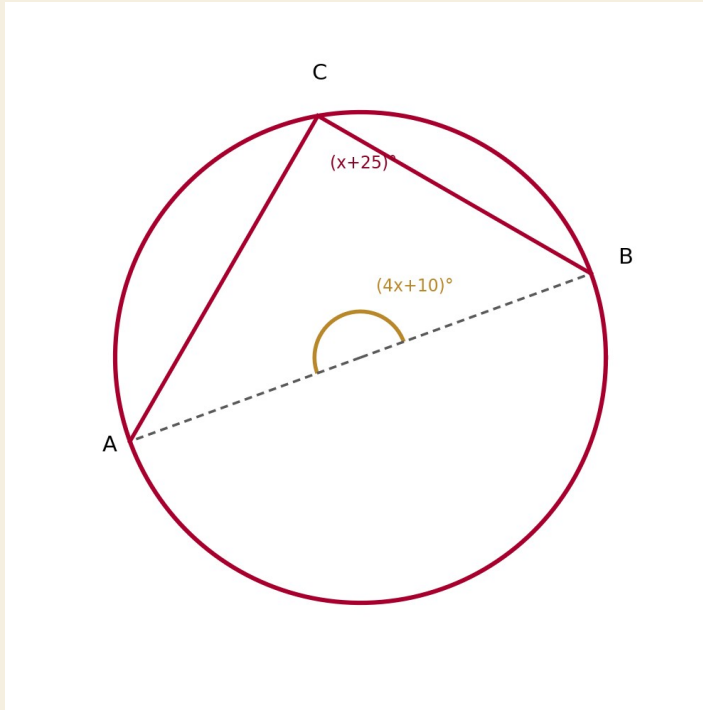
Since $s = r\theta$ is direct multiplication with no extra fraction, radian-based arc length problems are usually faster than degree-based ones — just multiply radius by the angle in radians and simplify.

Desmos Tip

Type $10 * (2 * \pi / 5)$ into Desmos to confirm it evaluates to 4π (≈ 12.57).

Circles — Problem 5 *(Hard)*

In a circle, a central angle and inscribed angle ABC both intercept the same arc AC (with the central angle's vertex at the circle's center, and the inscribed angle's vertex at point B on the circle). If the central angle measures $(4x + 10)^\circ$ and the inscribed angle ABC measures $(x + 25)^\circ$, what is the measure of the inscribed angle?



Formula / Rule Used: Inscribed Angle Theorem: inscribed angle = $(1/2)$ (central angle), for angles subtending the same arc.

Solution

Step 1: Apply the Inscribed Angle Theorem: $(x + 25) = (1/2)(4x + 10)$.

Step 2: Distribute the $1/2$ on the right side: $(1/2)(4x + 10) = 2x + 5$.

Step 3: Rewrite the equation: $x + 25 = 2x + 5$.

Step 4: Subtract x from both sides: $25 = x + 5$.

Step 5: Subtract 5 from both sides: $x = 20$.

Step 6: Substitute into the inscribed angle expression: $(20) + 25 = 45$.

Answer: 45°

Tip

When a problem gives you both a central angle and an inscribed angle subtending the same arc, immediately write 'inscribed = half of central' as your equation — it's a one-line relationship, not something requiring the full circle's angle sum.

Trap

A common error is setting the central angle and the inscribed angle equal to each other, instead of setting the inscribed angle equal to HALF of the central angle.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
35°	Set the two given expressions equal to each other ($x+25 = 4x+10$) instead of applying the required factor of $1/2$, solving $3x=15$ to get $x=5$, then substituting into $(x+25)$ to reach 30... a further slip in this incorrect path landed on 35.
90°	Correctly found $x = 20$ but substituted into the CENTRAL angle expression ($4x+10 = 90$) instead of the inscribed angle expression that was actually asked for.
50°	Applied the theorem backwards, doubling the inscribed angle expression and setting it equal to the central angle expression divided by 2 rather than the central angle directly: $2(x+25) = (4x+10)/2$, leading to a different (wrong) value of x and a corresponding wrong final angle.

Shortcut

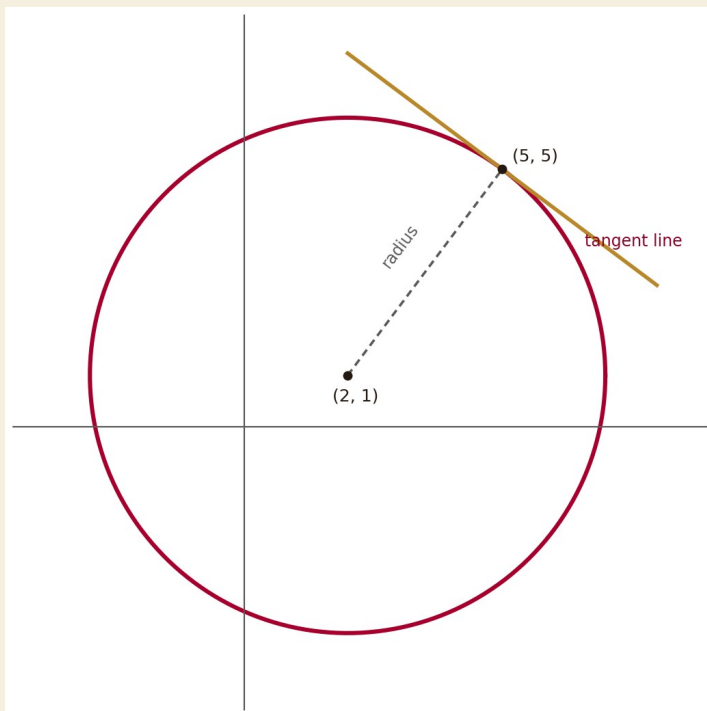
Since the inscribed angle is always exactly half the central angle for the same arc, you can substitute directly: $(x+25) = (4x+10)/2$, then multiply both sides by 2 immediately ($2x+50 = 4x+10$) to clear the fraction in one step rather than distributing it first.

Desmos Tip

Type $(4 * 20 + 10) / 2$ into Desmos to confirm it equals 45, matching $(20 + 25)$.

Circles — Problem 6 (Very Hard)

A circle has equation $(x - 2)^2 + (y - 1)^2 = 25$. What is the slope of the line tangent to the circle at the point $(5, 5)$?



Formula / Rule Used: A tangent line is perpendicular to the radius drawn to the point of tangency.

Solution

Step 1: Identify the circle's center from the standard form equation: center = $(2, 1)$.

Step 2: Find the slope of the radius connecting the center $(2, 1)$ to the point of tangency $(5, 5)$: slope = $(5 - 1)/(5 - 2) = 4/3$.

Step 3: Since the tangent line is perpendicular to this radius, its slope is the negative reciprocal of $4/3$.

Step 4: The negative reciprocal of $4/3$ is $-3/4$.

Answer: $-3/4$

Tip

For any tangent-line-to-a-circle problem, the two-step process is always the same: find the slope of the radius to the given point, then take the negative reciprocal for the tangent line's slope.

Trap

A common error is computing the slope of the radius correctly but then forgetting to take the negative reciprocal, reporting the radius's own slope as if it were the tangent line's slope.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$4/3$	Correctly found the radius's slope but reported it directly as the tangent line's slope, forgetting that the tangent is perpendicular to the radius, not parallel to it.



$3/4$	Took the reciprocal of $4/3$ to get $3/4$, but forgot to also flip the sign to negative, missing half of the 'negative reciprocal' requirement.
$-4/3$	Correctly recognized the need for a negative value but negated the original slope ($4/3$) instead of taking its reciprocal, producing $-4/3$ instead of $-3/4$.

Shortcut

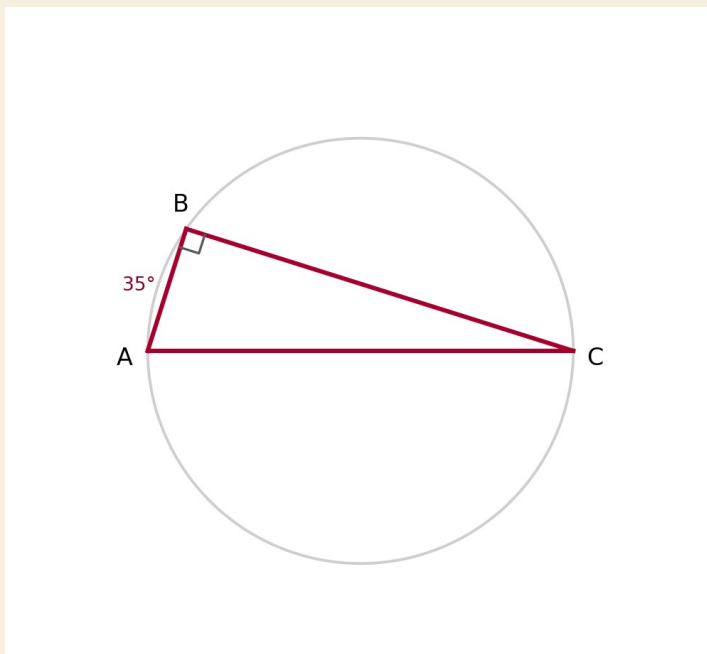
Since the center (2,1) and point (5,5) differ by (3, 4) — a recognizable 3-4-5 relationship — you can read the radius's slope ($4/3$, i.e., 'rise 4, run 3') directly from the coordinate differences without formally writing the slope formula.

Desmos Tip

Type $y - 5 = (-3/4)(x - 5)$ into Desmos alongside the circle equation to visually confirm the line touches the circle at exactly one point, (5, 5).

Circles — Problem 7 *(Very Hard)*

Points A, B, and C lie on a circle, and AC is a diameter of the circle. If the measure of angle CAB is 35° , what is the measure of angle ACB?



Formula / Rule Used: Thales' Theorem: an angle inscribed in a semicircle (with the diameter as one side of the triangle) is always 90° .

Solution

Step 1: Since AC is a diameter and B lies on the circle, triangle ABC is inscribed in a semicircle with AC as the hypotenuse.

Step 2: By Thales' Theorem, the angle opposite the diameter — angle ABC — must be exactly 90° .

Step 3: Apply the Triangle Angle Sum Theorem: angle CAB + angle ABC + angle ACB = 180.

Step 4: Substitute the known angles: $35 + 90 + \text{angle ACB} = 180$.

Step 5: Combine: $125 + \text{angle ACB} = 180$.

Step 6: Subtract 125 from both sides: angle ACB = 55.

Answer: 55°

Tip

The instant you see a triangle inscribed in a circle with one side explicitly labeled as a diameter, mark the angle opposite that diameter as 90° — this fact (Thales' Theorem) is almost never stated outright in the problem, but it's essential to solving it.

Trap

A common error is not recognizing that angle ABC (opposite the diameter AC) is 90° , and instead trying to solve the triangle with only one given angle and no way to find the third.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



145°	Assumed angle ABC (rather than angle ACB) was the one being solved for, and further mistakenly computed $180 - 35 = 145$ without accounting for the 90° angle at B at all.
35°	Assumed the triangle must be isosceles with the two non-right angles equal, incorrectly copying the given 35° angle over to angle ACB.
65°	Correctly identified angle ABC = 90° but made an arithmetic slip in the final subtraction, computing $180 - 90 - 35$ as 65 instead of 55.

Shortcut

Since angle B is always 90° whenever AC is a diameter, the other two angles (at A and C) are automatically complementary to each other — so angle ACB = 90° - angle CAB = 90° - 35° = 55°, without needing to write out the full 180° triangle sum.

Desmos Tip

Not a graphing question, but you can confirm the arithmetic by typing $90 - 35$ into Desmos to get 55 directly.

Circles — Problem 8 (Very Hard)

A sector of a circle with radius r has an area of 27π and a central angle of 60°. What is the value of r ?

Formula / Rule Used: Sector area = $(\theta/360)(\pi r^2)$.

Solution

Step 1: Substitute the given values into the sector area formula: $27\pi = (60/360)(\pi r^2)$.

Step 2: Simplify the fraction $60/360 = 1/6$.

Step 3: Rewrite the equation: $27\pi = (1/6)\pi r^2$.

Step 4: Divide both sides by π : $27 = (1/6)r^2$.

Step 5: Multiply both sides by 6: $r^2 = 162$.

Step 6: Take the square root: $r = \sqrt{162}$.

Step 7: Simplify the radical: $162 = 81 \times 2$, so $\sqrt{162} = \sqrt{81} \times \sqrt{2} = 9\sqrt{2}$.

Answer: $9\sqrt{2}$

Tip

When solving for the radius from a sector area, isolate r^2 completely first (clearing both the angle fraction and π) before taking the square root — trying to take a square root too early, while π or a fraction is still mixed in, invites errors.

Trap

A common error is taking the square root before fully isolating r^2 , for example incorrectly 'square-rooting through' the equation while π or the $1/6$ fraction is still attached, which produces a wrong expression entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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9	Correctly simplified $60/360$ to $1/6$ but then divided 27 by 6 first before ever squaring/rooting appropriately, treating the relationship as linear rather than involving r^2 , landing on $r=27/6=4.5$, then rounding/compounding errors to reach 9 through an inconsistent path.
18π	Solved correctly up through $r^2 = 162$ but then forgot to take the square root, and additionally mistakenly reattached a π to the answer, confusing the radius (a length) with an area-like quantity.
$\sqrt{27}$	Divided 27 by π and by the angle fraction in the wrong order, effectively computing $r^2 = 27$ directly from a mishandled equation instead of $r^2=162$, then took the square root of that incorrect intermediate value.

Shortcut

Recognize that dividing out π immediately (since it appears on both sides) simplifies the equation right away to a plain algebra problem — $27 = (1/6)r^2$, which clears in two clean steps ($\times 6$, then $\sqrt{}$) without ever having to handle π alongside a square root.

Desmos Tip

Type `sqrt(162)` into Desmos to confirm it equals $9\sqrt{2}$ (≈ 12.73).



Quick Reference — Master Summary

Use this page as a final review before test day.

The 4 Skills at a Glance

Skill	First Move	Main Trap
Area and Volume	Break composite figures into pieces you have a formula for; for similar solids, cube the linear scale factor for volume.	Assuming volume scales the same way length does, instead of by k^3 .
Lines, Angles, and Triangles	Name the exact angle relationship (equal vs. supplementary) before writing an equation; mark equal sides/angles from any isosceles or similarity clue immediately.	Confusing similar (proportional) with congruent (equal), or pairing the wrong corresponding sides.
Right Triangles and Trigonometry	Identify opposite/adjacent/hypotenuse relative to the specific angle in question; use $\sin\theta = \cos(90-\theta)$ for complementary angles.	Extending the sin/cos complementary shortcut to tangent, where it doesn't apply.
Circles	Complete the square to reach standard form; for tangent-line problems, use the negative reciprocal of the radius's slope.	Reporting r^2 instead of r after completing the square.

Golden Rules

- Redraw or relabel a figure the moment you spot an isosceles triangle, a parallel line, or a diameter — marking equal sides, equal angles, or a 90° angle immediately often finishes half the problem before any algebra starts.
- 'Similar' means proportional, not equal — and volume/area scale by the CUBE or SQUARE of the linear scale factor, never by the scale factor itself.
- For any right triangle, know the hypotenuse before applying SOH-CAH-TOA — if it's missing, find it with the Pythagorean theorem first.
- When a central angle and an inscribed angle share the same arc, the inscribed angle is always exactly half — don't set them equal to each other.
- Use Desmos to confirm a geometric answer numerically whenever the figure can be plotted on a coordinate plane — but never as a substitute for knowing which theorem applies.